The Economic Logic of China's Rise:

Geography, Big Push, and the Engineered Invisible Hand

April 2026

Calvin Chow
kickstar_chow@foxmail.com

The author's first language is Chinese. AI tools were used to assist with English expression throughout this paper. The ideas, arguments, and analytical framework are entirely the author's own, rooted in the experience of growing up in rural China. The core argument generates a testable prediction: if it survives empirical verification, it carries implications across multiple fields; if it does not, the framework is falsified. Preliminary empirical evidence, drawn from the broad correspondence between regional yield volatility and land-tenure patterns across Eurasia, suggests that the core prediction holds; however, precise confirmation awaits systematic quantitative testing with fine-grained agronomic and historical data.

Abstract

Surplus enables specialization; only stable surplus enables full specialization.

'The division of labor is limited by the extent of the market.' Traditionally, the extent of the market is analyzed through geographic reach or population size. Here we add a third dimension: the stability of the market over time.

More precisely, this article identifies geographic volatility, which determines the permanent, recurring volatility of grain output that climate imposes under pre-modern conditions, as a complementary dimension that refines the role of geography in development.

Specifically, building on Jeffrey Sachs's established perspective on geographic endowments, we propose a theoretical framework in which geography determines not only endowments but also volatility. The geographic endowments (soil quality, transport access, disease burden) shape the extent of the market; the geographic volatility shapes the reliability of price signals on which the market mechanism depends. By shaping the effective time orientation of individual agents and raising enforcement costs, this volatility determines the structural transaction costs of social coordination and shapes the boundaries of institutional evolution. Distributional institutional differences conventionally treated as causes of divergence, such as sharecropping versus fixed-rent tenancy or centralized versus limited government, are better understood as rational adaptations to different levels of volatility.

Crucially, while geographic volatility is an inherent natural constraint, modern technology and engineering can stabilize realized output. This framework thus accounts for the Great Divergence, when geographic volatility constrained institutions through volatile food supplies, and explains the heterogeneity of contemporary industrialization, where nations with limited capacity face higher de-risking costs. For late-developing nations, this framework implies that a Big Push is indispensable, but success hinges on whether it can achieve systemic agricultural de-risking.

Ultimately, the geographic volatility perspective reinforces Jeffrey Sachs's emphasis on the interplay of geography, technology, and institutions, strengthening the shared conclusion: geography precedes institutions, and the material foundation must be secured before markets can function.

Introduction: The Pivot of History

Today, China's industrial rise is the defining story of the global economy.

Western observers often view this shift simply as a threat to existing power structures or the resurgence of a fundamentally different civilizational model. Yet, these views miss the underlying logic: the central question is not just about geopolitical leadership, but about the principles by which humanity can survive and thrive under different natural constraints. Taking the long view of history, we see that China's path did not start today. It began with the civilization's distinctive foundational challenge: Faced with a land that was both immensely fertile and inherently volatile, how was this civilization to forge order amidst the chaos of nature?

This question points to a gap in economics and the social sciences more broadly. Traditional analysis has focused overwhelmingly on society itself (institutions, culture, policy) while paying insufficient attention to the relationship between humanity and nature. Jeffrey Sachs is a pioneering voice in economics: his work on the interplay of geography, technology, and institutions has forcefully demonstrated that geographic constraints precede institutional outcomes. Within this established perspective, the analysis has focused on the geographic endowments (soil quality, transport access, disease burden) that shape the level of output; and hence the extent of the market. This explains why some regions are poor (low output), but it struggles to explain why regions with high output, such as the Yangtze Delta, nevertheless failed to industrialize spontaneously.[1]

Industrialization means large-scale production: masses of people depend on the market to obtain food. This creates a critical requirement for the stability of grain output. It is precisely this requirement that points to a dimension that has remained underexplored as a determinant of divergent development paths: geographic volatility—the permanent, recurring volatility of grain output that climate imposes under pre-modern conditions.[2] By incorporating geographic

---

[1] Industrialization, as used in this article, refers to the transition from household-based production to a factory system characterized by concentrated production, systematic division of labor, and high fixed costs, not to mechanization as such. Adam Smith's pin factory, the founding image of political economy, demonstrated that the productivity leap of the factory is rooted in the division of labor, of which machinery is a later extension rather than a precondition. Marx later systematized this insight by distinguishing manufacture (centralized workshop production based on manual skill and division of labor) from modern industry (machine-based production); the former is already an industrial organization. The binding constraint identified by this framework, the vulnerability of any high-fixed-cost enterprise to volatility, applies equally to both. The reason traditional China never sustained large-scale workshops, let alone mechanized factories, is the same: recurrent shocks made any production system burdened by rigid overhead economically unviable. This also clarifies the relationship between this framework and two concepts in the existing literature. Mendels's "proto-industrialization" (1972) describes the expansion of rural household manufacturing for distant markets, while de Vries's "Industrious Revolution" (2008) describes the intensification of household labor and reallocation of time toward market-oriented production. Both phenomena were present in traditional China, yet neither led to the factory system. Within this framework, the explanation is straightforward: In high-volatility environments, household production is the rational organizational form precisely because it carries no rigid overhead and can retreat to self-sufficiency during downturns; proto-industrialization and the Industrious Revolution were not failed preludes to industrialization but stable adaptations to an environment that structurally excluded the factory form.

[2] In pre-modern agriculture, high variance and severe downside asymmetry are empirically bundled; this article uses 'volatility' as a shorthand for the compound risk profile, while noting the specific role of downside asymmetry where relevant.

volatility, we aim to offer a more complete account of how nature shapes the trajectory of human development. While the geographic endowments shape the extent of the market, the geographic volatility shapes the reliability of price signals on which the market mechanism depends. By shaping the effective time orientation of individual agents and enforcement costs, this volatility determines the structural transaction costs of social coordination and shapes the boundaries of institutional evolution.[3] Distributional institutional differences conventionally treated as causes of divergence, such as sharecropping versus fixed-rent tenancy or centralized versus limited government, are better understood as rational adaptations to different levels of volatility. As in biological evolution, adaptation to a shared environment constrains the viable set of institutional forms without uniquely determining which form emerges; within the feasible set, political agency, cultural inheritance, and historical contingency shape the specific outcome. This is not determinism but delimitation: the framework of this article does not explain why a particular institution originated or how it evolved, but it does explain why it persists—institutions endure because the material conditions that selected them endure.[4]

Specifically, in the primary comparison between China and the West, output levels are broadly comparable; the dimension along which the two decisively differ is volatility. Volatility directly shapes distributional institutions, the arrangements that determine who bears risk and who claims surplus, because it is precisely this question that recurrent output shocks make urgent. The institutional analysis throughout this article therefore focuses on how volatility constrains these distributional arrangements. This is a single dimension matched to a single class of institutions, not a claim that volatility explains all institutional variation. The Irish case, where stability coexists with lower output, tests the framework's capacity to distinguish the separate contributions of each dimension.

Crucially, while geographic volatility is an inherent natural constraint, modern technology and engineering can stabilize realized output. This distinction between the natural constraint and the engineered outcome provides the key to understanding both the Great Divergence and the heterogeneity of contemporary industrialization. In short, where England's invisible hand rested on providential stability, China's had to be engineered.

The article first develops the theoretical framework, showing how geographic volatility shapes rational choice, constrains the division of labor, and sets the material threshold of industrialization. It then applies this framework to the historical record—from England's providential stability and spontaneous industrialization, through traditional China's stagnation trap and contemporary China's engineered breakthrough, to the Soviet Union's cautionary failure. Ireland, where stability was decoupled from abundance, provides a supplementary test of the framework's explanatory power. The article concludes with policy implications for the Global South and a reflection on the renewal of civilization.

---

[3] Geographic volatility is critically binding because it recurs with every annual harvest, affects every person who must eat, and threatens the survival baseline on which all economic activity depends. No other source of volatility (war, epidemic, or commercial shock) shares this universality. See Appendix A for the full argument.

[4] What Douglass North called 'beliefs' or 'mental models' are better understood, within this framework, as primarily shaped by the physical environment rather than as independent priors that autonomously drive institutional choice. His 'path dependence' is likewise reinterpreted: institutions persist not because they are locked in, but because the material conditions that selected them persist. See Appendix B for the full argument.

Theoretical Framework: Complementing Endowments with Volatility

Industrialization means large-scale production: masses of people depend on the market to obtain food. This creates a critical requirement for the stability of grain output. It is precisely this requirement that points to the dimension of geographic volatility.

1. The Endowments-Volatility Framework

- **Geographic endowments:**
  - The established perspective attributes the failure of tropical or landlocked nations to low output or high costs (e.g., poor soil, expensive transport, high disease burden). This is a mean-based analysis, which accurately captures the development obstacles faced by many regions with harsh geographic conditions.
  - Theoretical limits: while this perspective effectively explains the poverty trap (e.g., in Sub-Saharan Africa), it struggles to explain the “involution trap” (the deadlock of self-sufficiency) as seen in traditional China. China's core regions possessed fertile soil and high yields. Contrary to assumptions of high transport costs, the Yangtze River system provided an extensive inland water transport network comparable to major European systems. Furthermore, located in the temperate and subtropical zones, it supported the world's largest population, indicating a low disease burden. Yet, this region nevertheless fell into stagnation.
- **Geographic volatility:**
  - The challenge for traditional China, under the continental East Asian monsoon, was not an inability to generate surplus, but the the permanent, recurring volatility that eroded it and destabilized the entire system built upon that surplus. Human activity can amplify this volatility, but the baseline of this volatility is climatically determined.
  - The destructive power of volatility: frequent floods and droughts (high volatility) mean that the process of capital accumulation is periodically interrupted or even reversed (a system-wide "reset"). The surplus created in bountiful years is often consumed or destroyed in lean years, locking society into a cycle of accumulation and dissipation. Beyond this direct destruction of capital, high volatility also disrupts price signals, shortens planning horizons, and raises enforcement costs, mechanisms we develop in the sections that follow. The core obstacle to industrialization lies not in the initial stock of capital, but in the stability required for capital to flow into long-term industrial investment.

2. Micro-Mechanism: From Geographic Volatility to Rational Choice

Geographic volatility does not dictate institutions directly; it shapes them by shaping the effective time orientation of individual agents, opening the path to development or confining a society to the imperatives of survival.

- **Stable environment (e.g., England):**
  - Price signaling: in stable environment, geographic stability acts as a natural insurer, ensuring that the price system provides clear and reliable signals.
  - Rationality: for market participants, this reliability lowers the discount rate, making a long-term orientation the rational default.
  - Agents' behavior: agents dare to specialize, trusting the market to supply what they no longer produce for themselves.
  - Capital behavior: capital flows into long-horizon, high-fixed-cost investments (e.g., tenant farming, factories, machinery) because a stable price environment ensures that residual can be effectively captured.
- **Highly volatile environment (e.g., Traditional China):**
  - Price signaling: here, frequent floods and droughts translate directly into persistent market noise and erratic price signals, rendering the price system periodically unreliable: while flexible, low-cost commerce could operate between shocks, recurrent disruptions prevented prices from serving as guides for long-term commitment.
  - Rationality: for market participants, this unreliability forces a high discount rate. Recurrent subsistence crises do not merely raise the required risk premium; they make near-term survival the overriding priority, rendering long-horizon payoffs effectively irrelevant. Short-term orientation becomes the rational default.
  - Agents' behavior: because the market cannot be relied upon across time, agents retain food self-sufficiency as a non-negotiable baseline, supplementing it with household crafts and occasional trade. This is not backwardness but rational diversification under endemic uncertainty.
  - Capital behavior: the factory system faces a structural barrier that makes large-scale production economically irrational: a factory requires high overhead, with rigid commitments to wages, rent, equipment, and debt service, implying high operating leverage. In an environment of recurrent shocks, these fixed costs become a structural liability: repeated demand slumps, typically triggered when spikes in grain prices drain purchasing power for industrial goods, progressively exhaust cash reserves and eventually lead to insolvency. In contrast, the small-scale household survives through its flexibility; with no rigid overhead, it can retreat to self-sufficiency during downturns that would destroy a larger enterprise burdened by fixed costs. Because industrial production is vulnerable to price noise and unpredictable shocks, its risk-adjusted returns are too low to justify the commitment of capital. Rational capital thus avoids the industrial sector, flowing instead into land, short-term lending, or hoarding, which effectively blocks industrialization.

3. Division of Labor: The Intertemporal Constraint

The micro-mechanism above allow us to refine Adam Smith's fundamental proposition: 'The division of labor is limited by the extent of the market.' Yang Xiaokai's inframarginal economics (Yang, 2001; Yang and Ng, 1993) extends and formalizes this proposition, showing that the

equilibrium level of specialization is endogenously determined by transaction costs. When transaction costs are high, agents rationally choose partial specialization or autarky; only when they fall below a threshold does full specialization emerge. Traditionally, the extent of the market is analyzed through geographic reach or population size. We add a third dimension: the stability of the market over time. Geographic volatility, we argue, is a binding constraint of transaction costs that has been insufficiently recognized.

Even if a market is spatially vast and integrated (like traditional China), it can be fragmented in the temporal dimension. Frequent natural shocks cause extreme price volatility, which periodically disrupts market transactions and breaks the continuity of the market. For the individual, the division of labor means specialization. **Surplus enables specialization; only stable surplus enables full specialization.** Where surplus is inherently unstable, rational agents do not fully specialize. They instead choose within-household self-sufficiency as a form of self-insurance. This constrains specialization, trapping the economy in a stagnant pattern of self-sufficiency.

4. Distributional Institutions: Context-Specific Adaptations

Based on these theoretical foundations, we must re-evaluate the distributional institutions that have conventionally been treated as causes of divergence.

- **Rethinking Land Tenure:**
    - The conventional view: sharecropping has long been criticized for its weak incentive properties (Marshall), as the tenant receives only a fraction of the marginal product, leading to suboptimal effort. Later refinements (Cheung, 1969; Stiglitz, 1974) reframe it as a trade-off between incentives and risk-sharing, but treat the level of output risk as an exogenous parameter, rarely tracing it to its physical origins or asking how its magnitude transforms the very nature of the problem (see Appendix C for a formal examination).
    - Our refinement: in highly volatile environments where output volatility is driven by severe climatic shocks, the problem is not merely balancing incentives against risk-sharing. Here the primary distributional challenge is the asymmetry of risk: while the upper bound of harvests in good years is relatively stable, the lower bound during disasters is unpredictable and often drops below subsistence levels. Under such conditions, rigid fixed-rent contracts become effectively unenforceable. Attempting to enforce a fixed payment during a major harvest failure does not lead to wealth accumulation, but to social friction, disputes, and potential violence. Because the risk of tenant insolvency is recurrent, landlords and tenants adopt sharecropping to provide the contractual flexibility necessary to maintain social order and long-term land productivity.
    - The Western counterpart: England's widespread use of rigid fixed-rent contracts has long been celebrated as institutional superiority; at a deeper level, it reflects the geographic stability that made such rigidity affordable. In a stable environment, the risk of a tenant being unable to pay is minimal, allowing the contract to be enforced as written with low enforcement costs.
- **Rethinking Property Rights:**

- The distributional logic of land tenure extends directly to the relationship between the state and the individual property owner. Sovereignty is the supreme ownership right over territory. The political community and the individual thus share ownership: the community retains a collective claim on output (taxation), and the individual holds the residual (property rights). This is not an analogy; it is the same distributional structure operating at a higher level. In a low-volatility environment, the tax base is predictable and transparent; the collective claim can be fixed, and the boundary between what belongs to the community and what belongs to the individual holds firm. This stability is what we call property rights protection. In a high-volatility environment, the tax base becomes both volatile and difficult to assess; the state must repeatedly intrude upon private holdings to extract resources, and the boundary becomes chronically unstable. Just as the rigidity of the English fixed rent was a dividend of geographic stability, not a superior invention, so too was the security of English property rights. In both cases, the same micro-mechanism is at work: output volatility determines the enforceability of fixed claims, at the private level (rent) and the sovereign level (taxation) alike.

- **Rethinking Kinship Networks and Social Structure:**
  - The rationality of kinship networks: this article does not ask how kinship systems originated or operated in detail; its concern is why strong kinship networks were the rational default in highly volatile environments. In this environment, where state-provided relief is insufficient and social order is unstable, the kinship system serves as the primary framework for risk-pooling and survival. Even in regions like the Yangtze Delta, where individual households may appear to function as nuclear families, agents still relied on extended kinship networks for mutual aid during shocks and collective security during crises. These institutions may well be cultural preferences; at a deeper level, they are rational adaptations to geographic volatility.
  - The Western divergence: symmetrically, in a stable environment, the natural predictability of output reduces the necessity of kinship-based protection. This enables a more atomized social structure, with the individual or nuclear family becoming the primary economic unit. This cultural preference for atomization significantly lowers the barriers to labor mobility. It enables individuals to seek specialized industrial work, free from the bonds of kinship.
  - Density versus pressure: population density and population pressure are distinct concepts. Density is a static measure: the number of people per unit of land. Pressure is a dynamic condition: the competitive squeeze of population against the carrying capacity of the land. The level of output determines the upper bound of density a region can sustain; volatility determines whether that density is channeled into specialization or locked into involution. Dense populations under low volatility can specialize, leave the land, and enter industry; the same density under high volatility is converted into pressure, as volatility ties labor to the land and incentivizes large families as self-insurance. What distinguishes involution from industrialization at high density is not the

number of people but the volatility of the output that sustains them.

- **Governance Paradox:**
  - This article sets aside the question of how different governance systems arose. What it addresses is why high-volatility environments systematically favor centralized states while low-volatility environments permit limited government. Both are understood as a class of governance arrangements, not a singular institutional design.
  - The structural contradiction: a highly volatile environment generates a high demand for state intervention (disaster prevention, domestic order, and external defense) while simultaneously undermining the fiscal capacity to fund it as the tax base becomes both volatile and difficult to assess. Faced with this paradox, states typically follow divergent paths: evolving into coercive fiscal states to survive, or suffering collapse due to fiscal failure.
  - The cost logic: volatility determines not which governance form a society chooses, but which forms are economically viable. In a high-volatility environment, the demand for state action (disaster prevention, famine relief, social order maintenance) is recurrent and urgent; a limited government lacks the capacity to meet it, and the cost of under-provision (famine, collapse) is existential. Governance is pushed toward centralization not by ideology but by the cost of the alternative. In a low-volatility environment, this demand is structurally low; centralized authority becomes more costly than it is worth, because the social costs it imposes (coercion, rigidity, political conflict) exceed the problems it exists to address. Through costly political experience, actors converge on limited government as the lower-cost equilibrium. The geographic lottery does not produce a specific governance form; it sets the cost structure within which human agency discovers viable arrangements.
  - The Western exception: England enjoyed a double blessing: geographic stability minimized the need for disaster prevention and lowered social friction, while the island geography reduced defense costs. This minimized the transaction costs of social coordination, making a limited government viable. In highly volatile environments, such a model would have been untenable.
- **Persistence and Transformation:**
  - The framework does not explain why a particular institution originated or how it evolved, but it does explain why it persists: institutions endure because the material conditions that selected them endure. This implies a corollary. When those conditions are fundamentally altered, whether by nature or by engineering, rational agents face a changed constraint set; their optimal choices shift, and institutions transform accordingly. An institution that successfully changes the conditions that selected it eliminates its own reason for being, not through failure, but through success. China's planned system arose because high output volatility rendered price signals unreliable to sustain full specialization; it then mobilized resources to engineer the very stability that nature had denied; once output volatility crossed the material threshold, a well-functioning market could take root on engineered stability, and the planned system negated itself

through the fulfillment of its own purpose. But before modern technology, this transformation was virtually unattainable. Pre-modern institutions could not alter the conditions that selected them, and those conditions persisted indefinitely. Traditional China's millennia of institutional continuity was not cultural conservatism; it was the persistence of the volatility that no pre-modern institution could overcome.

5. The Material Threshold of Industrialization

A clarification on the relationship between volatility and endowments is in order. The framework identifies volatility as the binding constraint on industrialization, but volatility does not operate in a vacuum: it must act upon an existing state. A minimum level of endowments, enough to sustain not merely a settled population but an organized political authority, is the precondition for any of the framework's mechanisms to operate. Below this threshold, only stateless societies exist, and the institutional infrastructure that volatility would constrain is absent. This is the domain addressed by Jared Diamond, and it corresponds to the Absent or Paper Leviathan identified in the governance analysis. Once a state exists, volatility becomes the binding constraint: a low-volatility society will gradually deepen its division of labor and develop factory-scale organization, because the high-fixed-cost structures that define industrialization become economically viable wherever demand is predictable and contracts enforceable. This process does not require steam power, cheap coal, or navigable rivers. It requires only that surplus be reliably sustained. The additional endowments that England enjoyed, abundant coal, dense waterways, and island security, determined the specific form and extraordinary speed of the Industrial Revolution, but they were accelerators, not prerequisites for industrialization as such. Ireland confirms the distinction: lacking coal, burdened by poor internal connectivity, yet possessed of low volatility, it developed factory-scale agro-processing and manufacturing over the course of a century, slowly but irreversibly. Endowments and volatility thus operate on different logical levels: endowments set the floor for state existence and the ceiling for speed; volatility determines whether the path between floor and ceiling is traversable at all.

In summary, industrialization is not just a matter of capital accumulation, technological innovation, or the design of incentive mechanisms; it is, at its core, a process of de-risking. Only when output volatility is tamed can a well-functioning market take root, allowing capital to accumulate, technology to advance, and incentives to function as designed.

- **The Western path (providence):** the West benefited from a providential environment that provided both favorable and predictable grain output. This was the binding advantage: while other factors (transport, energy, geopolitical security) also mattered, it was agricultural stability that allowed well-functioning markets to emerge and industrialization to unfold spontaneously.
- **The Chinese path (engineering):** faced with inherently high grain output volatility, a purely market-based system would struggle to sustain the intertemporal stability required for industrialization. This demanded a proactive state to undertake systemic engineering to mitigate the effects of geographic volatility. Through these material investments, the state engineered the conditions for a well-functioning market and a

subsequent industrial takeoff.

6. Global Perspective: Industrial Takeoff & The Geographic Lottery

Endowments, volatility, Institution, and industrialization across the West, Russia, and China.

	Endowments-Volatility (19th Century)			Baseline Institution	Threshold
Regions	Transport Connectivity	Grain Yield	Geographic Stability	Land tenure	Industrial Takeoff
Yangzi Delta China	5	4-5	2	sharecropping	1950 (Big Push)
Japan West	4-5	3-4	4	fixed-rent	1880
Japan East	4	3-4	3	complex	1900
Ireland North	4	3	5	fixed-rent	1840
Ireland South	3	3	5	fixed-rent	1960
England	5	3-4	5	fixed-rent	1780
American North	4	2-3	4	fixed-rent	1830
France North	4	3-4	5	fixed-rent	1830
Netherlands	5	3-4	5	fixed-rent	1860
Belgium	5	4-5	5	fixed-rent	1820
Germany North	4	2-3	3-4	fixed-rent	1840
Germany South	3	3-4	3-4	fixed-rent	1880
France Central	3	2-3	3-4	complex	1890
Italy North	5	3-4	3-4	complex	1890
Spain Industrial	3-4	2-3	3-4	complex	1840
American South	3-4	2-3	2	sharecropping	1950
France South	3	2-3	2	sharecropping	1950
Italy South	3	2	2	sharecropping	1950
Spain Agricultural	2-3	1-2	2	sharecropping	1950
Greece	3	1-2	2	sharecropping	1950
Ukraine (USSR)	3-4	3	3	complex	1930 (Big Push)
Russia (USSR)	1-2	1-2	1-2	labor-rent	1930 (Big Push)

Table Note: Transport Connectivity, Grain Yield, and Geographic Stability are rated on a relative scale from 5 (Highest) to 1 (Lowest), intended for cross-regional comparison rather than absolute measurement. Ranges (e.g., 2-3, 3-4) are allowed to reflect intra-regional and temporal variation. Geographic Stability refers to the interannual volatility of grain yields under prevailing climate. "Complex" land tenure indicates a mix of fixed-rent and sharecropping. This table illustrates the broad correspondence between geographic dimensions and the timing of industrialization.

A final observation from this table puts the framework in global perspective.[5]

[5] The ratings in this table are qualitative ordinal scores. Each column draws on an independent body of established knowledge: Geographic Stability on the spatial distribution of precipitation variability documented in

The low-volatility conditions that enabled spontaneous industrialization are not the norm on Earth's surface but a narrow geographic exception. In the North Atlantic zone, oceanic currents and prevailing westerlies deliver rainfall with unusual regularity, creating a slender band of maritime-temperate stability: the American Northeast on the western shore of the Atlantic, and southeastern England, the Low Countries, and northern France on the eastern shore. These regions share a further advantage in their relatively high latitude, which substantially attenuates the tropical variability driven by ENSO cycles that destabilizes lower-latitude climate systems. Moving in any direction from this core, volatility rises: southward into the Mediterranean's seasonal aridity, eastward into continental interiors with their increasingly erratic rainfall, and deep into the monsoon zones where the timing, spatial distribution, and intensity of water are subject to large interannual variation. Maritime Japan stands as a partial exception at the eastern edge of Eurasia. The archipelago is bathed by the Kuroshio and Tsushima warm currents, whose thermal influence penetrates a landmass narrow enough that no point lies far from the sea. Its core agricultural regions lie well south of the North Atlantic core, yet far enough from the tropics that the direct influence of ENSO-driven variability is substantially attenuated. Together, warm currents, narrow landmass, and sufficient latitude moderate the monsoon extremes that define continental East Asia, producing a climate that, while more variable than the North Atlantic core, rainfall remains strongly influenced by the baiu front and typhoons, is far more stable than the adjacent mainland.

Beyond these limited zones of stability, the vast majority of the world's agricultural civilizations, including China, India, the Middle East, the Mediterranean world, Russia, and Sub-Saharan Africa, have always operated in high-volatility environments. This reverses the default assumption of traditional institutional economics: spontaneous industrialization is not the norm from which most societies deviated, but a narrow geographic exception. The vast majority of the world's agricultural civilizations have always operated in high-volatility environments where spontaneous industrialization was structurally precluded. The few societies that did industrialize spontaneously happened to sit in zones of exceptional geographic stability, whether provided by Atlantic westerlies or by oceanic encirclement. Where nature provided stability, a well-functioning market could emerge spontaneously, and Adam Smith's invisible hand operated on a providential foundation. Where nature did not, the foundation itself had to be built before a well-functioning market could take root. The "engineered invisible hand" is not a deviation from the normal path; for most of humanity, it is the only path available.

7. Empirical Basis and Testable Predictions

The geographic patterns of climate-driven agricultural volatility are stable at centennial to millennial timescales. The monsoon regime that shaped traditional China's output volatility persists today, as does the maritime stability that favored northwestern Europe. Modern

---

climatological datasets (e.g., CRU, GPCC); Transport Connectivity on physical geography (distance to coast, navigable waterway density, terrain) directly readable from the landscape; Grain Yield on the systematic historical yield reconstructions of agricultural historians (Slicher van Bath, Allen, Overton, Li Bozhong, among others); Land Tenure on the extensive regional agrarian history literatures; and Industrial Takeoff on consensus periodizations in economic history.

climatological and agronomic data on interannual yield volatility therefore provide a reliable empirical foundation for the historical volatility that this framework invokes. More precisely, modern meteorological records, fed into low-input agronomic models that strip out the effects of modern irrigation, fertilizer, and improved seeds, can simulate the yield distributions that pre-modern farmers would have faced, providing a direct empirical window into historical volatility.

A particularly tractable test lies in the relationship between interannual yield volatility and land tenure. The framework predicts a threshold in the coefficient of variation of grain yields (the ratio of standard deviation to mean): regions below it should exhibit rigid fixed-rent contracts, regions above it should exhibit sharecropping or flexible rent-abatement practices, and regions near it should exhibit mixed arrangements. Preliminary indications from modern yield data suggest that the boundary separating historically fixed-rent from sharecropping regions falls in the range of roughly 12 to 20 percent. Since modern agricultural inputs have compressed yield volatility, particularly in formerly high-volatility regions, the pre-modern gap between fixed-rent and sharecropping zones was likely wider, and the threshold correspondingly sharper. Modern climatological and agronomic data on yield volatility across the historical grain-producing regions of Eurasia can provide the quantitative test, though precise identification awaits systematic empirical investigation.[6]

8. Scope: What the Framework Explains and Does Not Explain

This article focuses on the direction of selection pressure that volatility imposes on distributional institutions: the arrangements that govern how output and risk are shared among agents, such as land tenure, kinship structures, and governance forms. These are the institutions most directly shaped by the volatility of output, because they answer the question that volatility makes urgent: who bears the risk and who claims the surplus.

The same volatility conditions also shape closely related phenomena that are not institutions in the strict sense but are inseparable from the distributional logic: population strategies adopted under endemic uncertainty, the direction of capital flows between sectors, the economic viability of labor-saving technology, and the material prerequisites of monetary stability. The historical

---

[6] Two clarifications are in order regarding the relationship between variance and downside asymmetry, and the implications of human amplification for empirical testing.

First, in pre-modern agriculture, high variance and severe downside asymmetry are empirically bundled: the more climatically unstable a region, the greater the frequency and severity with which disaster-year output falls below subsistence. The coefficient of variation therefore serves as an adequate proxy for the compound risk profile in threshold testing. The simulation method itself generates the full yield distribution, including the shape of the left tail; the coefficient of variation is a convenient summary statistic extracted from that richer output, not an information constraint.

Second, under pre-modern conditions, the human amplification of volatility operates as a positive feedback: in high-volatility environments, human responses (hillside clearing, lake reclamation) intensify volatility, while in low-volatility environments, the impact of human activity on volatility is comparatively minor. This preserves the cross-regional ranking: actual pre-modern volatility is higher than the climatic baseline in high-volatility regions and close to it in low-volatility regions. Coefficients of variation derived from climate data fed into low-input agronomic models therefore do not invert the regional ordering and do not compromise threshold verification.

analysis treats these as extensions of the distributional core rather than independent explanatory targets.

Innovation of all kinds, technological, scientific, and institutional, requires only surplus and the division of labor it enables. Any civilization that generates surplus produces such innovations: China invented the water-powered grand spinning wheel and developed the imperial examination system, the Islamic world developed algebra and the commenda partnership, ancient Greece formalized geometry and democratic self-governance. Many of the institutional forms celebrated as foundations of Western modernity have parallels in these earlier traditions. What differed was not the capacity for invention but the conditions for sustained accumulation. Traditional political economy already accounts for the link between surplus, division of labor, and innovation. What it leaves unanswered is why some innovations accumulated into transformative systems while others flashed across the sky and vanished. The specific origins of certain innovations of major consequence, such as the autonomous university or the legal protection of intellectual property, involve cultural traditions, intellectual lineages, and historical contingencies that this article does not attempt to explain. What it does address is the material conditions that determined whether innovations, once they appeared, could survive, accumulate, and produce systemic consequences.

More broadly, the framework identifies the boundaries of the feasible set of distributional institutions that a given material environment can sustain. It does not explain what happens inside that set. The following fall outside its scope:

- How specific institutional forms are selected, designed, or contested within the feasible set. Political agency, cultural inheritance, and historical contingency shape the particular path taken among viable alternatives, in ways that the framework does not derive. It explains why low-volatility environments favor rigid contracts and limited government, but not why England arrived at parliamentary monarchy while the Netherlands developed a republic; equally, it explains why high-volatility environments systematically favor centralized states, but not why China developed the examination-based civil bureaucracy while the Ottoman Empire developed the devshirme military-administrative system.
- How specific social structures such as kinship systems or nuclear families originate and operate in detail. The framework predicts that high volatility favors extended kinship networks and low volatility favors nuclear families, but the internal organization of the Chinese lineage system, the European manor, or the Japanese ie household involves cultural, religious, and legal specificities beyond its reach.
- What accounts for timing differences among regions that share similar geographic conditions. Why did Belgium industrialize in the 1820s while the Netherlands waited until the 1860s? These differences may reflect political, commercial, strategic, and other contingencies whose effects lie outside this framework.
- How concentrated power structures manage or fail to manage the systemic risks inherent in operating with razor-thin margins. The framework identifies why such catastrophic episodes were structurally possible, but the specific political dynamics that produced them belong to the study of centralized governance, not this framework.

- How governance modes concretely transition as engineered stability shifts the balance from transformation toward preservation. The framework explains which institutional forms the altered conditions select for; the competitive process through which agents discover these forms is no different from that which produced England's constitutional settlement. The specific sequencing of reform, the role of intra-elite contestation, and the pace of legal development are questions of political process that this framework does not address.

These exclusions share a common character: they are questions about variation within the feasible set, whereas this article's subject is the boundaries of the set itself.

Prologue: Law and Engineering

At the formative stage of their respective civilizations, two legendary events set the course for the divergent trajectories of governance in the West and China.

The first centers on Mount Sinai, where Moses ascended alone to receive divine revelation. He brought down not food, which was already a providential gift, but the Ten Commandments. In the wilderness, Moses established order through **"The Law"**: a covenant between man and God, and by extension, between people. This represents the preservation-oriented governance of the West, where order is seen as a divine inheritance. Centered on the preservation of eternal rules, it presupposed a well-provisioned sanctuary—an 'oasis' where the primary governance task is settling disputes about rights. In the modern West, this has evolved into a deep reliance on the rule of law.

The second unfolds at Mount Tu. Here, Yu the Great, the legendary sage-king and founder of China's first dynasty, gathered the tribal leaders, but only after thirteen years of wading through mud to tame catastrophic floods. Faced with nature's chaos, he did not wait for divine intervention. Instead, he mobilized the people to dredge rivers and level hills, physically reshaping the landscape. Yu the Great established order through **"Engineering"**: transforming nature to define the realm. This represents the transformation-oriented governance of China, where mastery over environmental volatility provides the essential platform upon which social order and rules of governance can subsequently be built. In modern China, this logic is manifested in an engineering-led state.

For centuries, the rise of the West persuaded the world that rights-focused governance was the only path. Today, China's successful Big Push reasserts an alternative logic of engineering-led governance.

Historical Analysis

The historical analysis that follows applies the endowments-volatility framework to the comparative record. Where institutional differences are discussed, the focus is on distributional institutions and their extensions: the region-specific arrangements through which societies allocate risk and surplus, such as land tenure, kinship structures, and governance forms, together with the closely related phenomena of population strategies, capital flows, technological viability, and monetary stability that the same volatility conditions shape.

I. Geography and the Survival Challenge

At its core, human history is the story of how people respond to nature's challenge of survival and development. Different geographic environments set different rules for survival, shaping distinct civilizations.

Chinese civilization was born from this very challenge. Its land is far from barren; instead, it possesses vast, fertile plains that have supported the world's largest population. For most of history, this natural wealth allowed China to maintain the world’s largest economy.

Yet, this abundance came with a dangerous catch: high geographic volatility. Driven by the continental East Asian monsoon, the unpredictable timing and volume of water translate into significant fluctuations in grain output. Rarely has a civilization been so defined by its struggle against nature, yet endured for millennia. From the Dujiangyan irrigation system to the Grand Canal, China's engineering miracles were essential material investments to survive in a highly volatile environment.

This constant struggle reinforced a tradition of practicality over dogma. In the face of ruthless natural laws, empty rhetoric offers no protection. Survival required massive organization and physical engineering. In China, survival was never a providential gift; it was an outcome earned through persistent effort in the face of recurring crises.

Nevertheless, while this immense survival pressure stimulated China's potential to transform nature, it also built an invisible wall. The constant struggle just to survive locked the society into a stagnation trap, blocking the spontaneous path to industrialization.

When the Industrial Revolution arrived, the glory of this agrarian civilization suddenly became a liability. An agricultural nation at the mercy of the monsoon could not withstand an industrial power armed with machines. This, fundamentally, is the essence of China's "century of humiliation"—not a cultural failure, but the collapse of a pre-industrial society facing industrialized military powers.

To truly understand why industrialization failed to emerge spontaneously in China, we must first dispel a common misconception. Historians often blame this stagnation on overpopulation or coercive institutions. But these are symptoms, not causes.

Overpopulation was not an independent cause of stagnation but was itself driven by volatility: in a highly volatile environment, a large family was the primary mechanism of self-insurance against crisis. Likewise, an intrusive state was not an arbitrary imposition but a response to the governance paradox: in a highly volatile economy, the immense difficulty of reliably monitoring and collecting revenue forced the government to be coercive in securing its fiscal base.

Volatility is the deeper cause; population pressure and institutional forms are primarily adaptations to it, not independent sources of stagnation, even though under pre-modern conditions they amplified the very volatility that shaped them.[7] We can only understand the true threshold of industrial civilization by grasping this reality.

II. Foundations of a Well-Functioning Market

Industrialization does not happen in a vacuum; it requires specific social and economic preconditions. These conditions are not providential gifts, except for a few lucky nations. The most critical prerequisite is the formation of a well-functioning market. In economic theory, a well-functioning market is defined by its ability to coordinate resources through reliable price signals and predictable rules. While mainstream economic models routinely assume the clarity of price signals (and even where the literature examines signal distortion, it typically attributes the problem to information asymmetry between agents), the role of the physical environment itself as a structural source of price instability has not been adequately incorporated into the theoretical frameworks that underpin development economics. This is precisely where theory parts from reality.

A well-functioning market does not appear simply by adopting a constitution or hanging up a "free market" sign. Its prerequisite is stable access to essential material supplies: the clarity of price signals, and even the predictability of rules, ultimately rest upon material fundamentals. In pre-modern societies, food was the essential supply. In the modern era, energy has joined it.

---

[7] Traditional China did develop mechanisms to buffer volatility, including state granaries, long-distance grain transport, and kinship-based mutual aid, but these were palliatives operating within the constraint, not solutions that removed it. Long-distance grain shipment, epitomized by the Grand Canal, could redistribute surplus between regions, but each major grain-producing region spanned vast areas within which climatic shocks were highly correlated; pre-modern transport capacity could not compensate for the deficit when an entire region failed simultaneously. Moreover, monsoon-driven disasters sometimes clustered across consecutive years, rapidly exhausting accumulated reserves and overwhelming any buffering mechanism. Even when the climate returned to normal, the effects of a single major disaster extended well beyond the initial shock: damaged infrastructure, depleted granaries, lost livestock, and cascading afflictions such as locusts and epidemics prolonged the crisis for years. In an environment where severe shocks recurred every few years, the next disaster often arrived before recovery from the last was complete, creating a compounding cycle that no pre-modern buffering mechanism could break. Whether administered by the state or organized by kinship networks, none could eliminate the systemic volatility that only large-scale engineering would eventually suppress.

Across all eras, however, the supply of food remains the very foundation of economic life. The volatility of grain output determines the stability of market. As Emperor Qianlong observed, grain prices set the anchor for all other prices. When this anchor holds firm, prices transmit clear value signals rather than survival panic, and contract enforcement remains straightforward. This provides the stable expectations required for specialization and long-term commitments. In contrast, in a highly volatile environment, price signals are clouded by recurrent shocks, and contract enforcement grows costly as defaults become endemic; stable expectations are destroyed, rendering full specialization and long-term commitments unviable. In this sense, the "invisible hand" of the market is only effective when it rests upon the solid ground of material predictability. Without this stability, all rational individual decisions point toward hedging for survival, effectively blocking industrialization.

III. The Western Baseline: Providential Stability

The rise of the West was multidimensional, but fundamentally, it rested on the overwhelming material power generated by the Industrial Revolution. England, which nurtured this revolution, was a special case where a well-functioning market formed and industrialization unfolded spontaneously. However, this owed less to abstract culture or institutions than to a geographic jackpot: a rare case where agricultural stability, energy abundance, easy transport, and strategic security converged. While other regions had their own resources, England enjoyed the providential luxury of predictability—a decisive advantage that required no prior physical engineering.

1. The Geographic Jackpot

England possesses a mild and moist maritime climate. Influenced by the North Atlantic Drift and westerlies, rainfall is reliable year after year and stays remarkably even throughout the seasons. This favorable stability is exceptional. Just across the sea, Ireland and Western Scotland suffer from excessive rainfall that makes wheat farming marginal; to the south, Mediterranean France experiences higher volatility; to the east, the North German Plain contends with harsher winters. But Southeastern England, along with the Low Countries and northern France, sits in a perfect "Goldilocks zone."[8] In sharp contrast to the volatile monsoon climate of China, English farmers were spared the violent cycle of feast or famine. While mean yields might not have matched the intensive rice paddies of East Asia, the volatility, the binding constraint, was exceptionally low.

---

[8] The Low Countries are famous for their hydraulic engineering, but the challenge it addressed was fundamentally different. Dutch water threats, principally storm surges and river floods, came from known directions and could be countered with known methods: dikes, polders, and windmill-driven drainage solved the problem within the reach of pre-modern technology. The monsoon-driven drought-flood alternation that defined Chinese agriculture was not similarly tractable: drought has no direction against which a dike can be built, and basin-wide flooding cannot be intercepted at any single point. No pre-modern engineering could suppress it at the systemic level. More broadly, the Netherlands is often understood as a case driven primarily by commercial prosperity. This is accurate but incomplete. The Dutch maritime climate delivered a low-volatility baseline comparable to southeastern England; fixed-rent tenancy predominated. The advanced commercial farming, together with fisheries and overseas trade, sustained the long-horizon investment on which Dutch prosperity depended. The Netherlands shared England's geographic stability; but limited territory and greater exposure to geopolitical threats made it a smaller, more vulnerable England.

It was this natural stability in grain production that allowed England to embark on a path different from that of the vast continental and monsoon regions of Eurasia. Here, governance could focus primarily on delineating boundaries and writing leases. Grain harvest was reliable; effort translated directly into reward. Even England's most contentious agrarian transformation, the enclosure movement, was driven by the prospect of long-term returns on land consolidation and improvement — not by harvest failure or famine. Unlike the Yangtze Delta, where stable yields required enormous investment in irrigation, drainage, and flood control, England's agricultural stability demanded minimal hydraulic engineering—a natural gift that delivered a better result without the crushing cost.[9]

2. Fixed-Rent Tenancy and Predictability

To understand the rise of the West, we must look closely at the subtle but decisive difference in the baseline institution: land tenure.[10] While economic historians often note that fixed rent (paying a set amount regardless of harvest) is the hallmark of modern commercial farming, this institution was not unique to England. Since fixed rent appeared independently in multiple civilizations, the key question is not why it was invented but why the same institution produced transformative results in some environments and not in others.

In advanced agrarian regions like the Yangtze Delta in China, fixed rent (known as *iron-rent*) was a common legal practice, and perpetual tenancy (known as *yongdian*) also existed in certain localities. The legal form was nominally the same; it was the physical volatility of the environment that determined how the contract operated in practice.

Risk-sharing in China: A Flexible Survival Mechanism

Under the continental East Asian monsoon, where floods and droughts were frequent, a rigid fixed rent was effectively unenforceable. If strictly implemented, a bad harvest would lead to mass tenant default and violent uprisings. Therefore, Chinese society evolved rent abatement

---

[9] England did undertake significant hydraulic works, most notably the Fenland drainage in East Anglia, which expanded arable land and also managed local flood risk. But England's maritime climate delivered baseline agricultural stability across the vast majority of its territory without hydraulic intervention. These projects were enhancements, not prerequisites for a functioning agrarian order. In China, by contrast, system-wide engineering across entire river basins was indispensable merely to suppress the flood-drought volatility that repeatedly threatened social order. England also experienced dearth years beyond the three major crises of 1315–17, the 1590s, and the 1690s. But apart from 1315–17, these were predominantly price events within a functioning social order, qualitatively different from the basin-wide shocks in China that triggered mass displacement and dynastic upheaval.

[10] We call land tenure the 'baseline institution' because it occupies a unique position in the institutional hierarchy: it is the interface through which the physical constraints of nature are most directly translated into distributional arrangements, determining who bears the risk and who claims the surplus of agricultural production. In agrarian societies, land tenure governed the distribution of the primary source of wealth, and the arrangements it fostered shaped the evolution of other distributional institutions from contract law to state finance. Land tenure is indeed the oldest institution to have accompanied civilization since its origin, but its primacy is not merely historical. Agriculture is not merely the starting point of civilization but its permanent foundation. Even in industrial societies, the entire edifice of specialized production rests on a premise that must be continuously satisfied: participants in the division of labor do not need to grow their own food. When that premise fails, all higher-order institutions collapse. Land tenure is 'baseline' because it governs this foundation, and will remain so as long as human survival depends on a stable food supply.

practices, sometimes tacit custom, sometimes mandated by official policy, for times of famine. This was not a Chinese peculiarity but a practice widely attested across pre-modern agrarian legal traditions: Roman jurists had long codified the principle of *remissio mercedis,* rent abatement for harvest failure caused by forces beyond the tenant's control, on the ground that no contract can justly bind a party against the force of nature. Whenever disaster struck, tenants had the right to demand a rent reduction. Consequently, the fixed rent was nominal. In reality, it constantly changed with the fluctuations of nature, effectively reverting to a form of sharecropping, where risks were shared. Furthermore, perpetual tenancy (*yongdian*) was a legal defense against an environment that constantly threatened to dispossess the tenant, yet even this legal shield crumbled when the survival crisis forced tenants to sell their rights just to buy food. Legal stability collapsed under economic volatility. The contract was not an anchor for capital accumulation, but a flexible buffer for tenant survival.

Rigidity in the West: The Engine of Development

England, however, enjoyed the geographic jackpot of stability. Here, the English fixed rent was a different species: it was genuinely rigid. Because extreme harvest failures were rare, famine abatement clauses were largely absent from the system. Contracts were enforced as written. Where abatement occurred, it was either stipulated in individual leases or granted voluntarily by landlords. It was not a systemic legal principle, but an occasional private arrangement that the low frequency of harvest failure made rarely necessary. This rigidity, seemingly a harshness, was actually a blessing. It meant that the tenant farmer, after paying the fixed rent, became the residual claimant of all surplus. And tenure security followed the same logic. In England, the standard arrangement was not perpetual tenure but fixed-term leases—typically of 7 to 21 years. No legal guarantee of permanence was necessary in practice: low output volatility ensured that tenants reliably paid their rent and landlords reliably renewed the lease. Tenure security was a dividend of geographic stability rather than solely a legal invention. Predictability is the foundation of long-term investment and capital accumulation: Knowing that "nature will not steal my surplus," English farmers dared to invest heavily in land improvement and machinery while retaining the gains year after year. Thus, capital accumulated. The English fixed rent was less a superior invention of human wisdom than a privilege of predictability that a stable environment made affordable.[11]

The Exception that Proves the Rule: Dujiangyan and the Po Valley

The case of the Dujiangyan irrigation zone on the Chengdu Plain, built in the third century BC and still operational today, provides a crucial exception that confirms this risk-based logic. In this local oasis where engineering mastered the natural volatility, the fixed rent (*iron-rent*) indeed became as genuinely rigid as its English counterpart. Because the risk of harvest failure was virtually eliminated, land contracts in this region achieved genuine rigidity: famine abatement clauses were either absent or, where present, rarely invoked in practice. It fostered a wealthy class of managerial farmers, proving that Chinese agents were fully capable of developing rigid contractual institutions once the physical environment allowed for it. However, this local success failed to ignite a national industrial revolution. Such stable zones were extraordinarily scarce and

[11] The apparent independence of the parameters in Stiglitz's (1974) model proves, under the empirical conditions of pre-modern agriculture, to be an artifact of the formalization. The effective degrees of freedom reduce to one: the magnitude of output volatility. See Appendix C for the full argument.

small. They simply could not overcome the overwhelming burden of continental volatility. A comparable process occurred in the Po Valley of Northern Italy. By the late 19th century, its core engineered stability zone spanned approximately 10,000 square kilometers. This scale fostered similar fixed-rent contracts and managerial farming modes across a vast contiguous belt. At the turn of the 20th century, this zone was large enough and sufficiently connected to Mediterranean trade to accumulate the capital and demand necessary for Italy's northern industrial takeoff. In contrast, the Dujiangyan core remained a geographically isolated sanctuary of about 2,000 square kilometers. Hemmed in by the mountains of the Sichuan Basin, it remained a "brilliant but isolated dot" on a vast, volatile continent.

3. Rise of Tenant Farming

Built upon this foundation of rigid, predictable fixed rents, England developed the tenant-farming system to its fullest extent. As an operator, the tenant farmer rented land at a fixed rent, hired laborers at fixed wages, and retained the residual. This business model was the precursor to the modern enterprise, which follows the same logic: paying fixed interest to banks, fixed wages to workers, and claiming residual profit. The point here is not to trace the origins of the enterprise form, but to show why this fixed-cost structure was economically viable in a stable environment and not elsewhere.

This structure implies high operating leverage. For a business burdened by rigid fixed costs (fixed rent and wages), revenue stability is a matter of life and death. In highly volatile environments, the challenge is not just an occasional bad harvest, but structural volatility. The incessant swings between feast and famine make future cash flows impossible to predict. Under such uncertainty, adopting a high-fixed-cost structure is economically unviable; recurrent lean years would eventually drain liquidity and bankrupt the operator.

This explains why large-scale commercial farms failed to emerge in traditional China despite high land productivity. In England, however, nature acted as the powerful insurer, smoothing revenue streams and allowing this high-leverage model to thrive. While family farms persisted and scale had its limits, the dominant trend was the consolidation of land into efficient, capital-rich units. Thus, tenant farming gradually grew and prospered, turning agriculture from a survival activity into a capital-accumulating industry.

While tenant farming was England's distinctive path, the same underlying logic operated across the stable zone in varied forms. In the Paris Basin, large fixed-rent farms closely mirrored the English model. In the Low Countries, monetized fixed-rent tenancy predominated, embedded in a highly commercialized economy. In the American North (New England and the Mid-Atlantic), independent freeholders were the norm, though tenancy was not absent. The specific agrarian forms varied, but the common thread was the viability of fixed-cost arrangements in a stable environment: obligations could be met reliably, and a substantial share of the surplus could be retained and reinvested.

4. The Foundation of Social Coordination

The natural stability of grain output brought profound social consequences. Adam Smith, drawing on the English experience, proposed three elements for wealth growth: "Peace, easy taxes, and a tolerable administration of justice." For centuries, this has been viewed as the universal institutional recipe for prosperity. But in reality, these are conditions that geographic stability made possible:

- **Peace**

  Not just the absence of war, but low internal social friction. When the future is predictable, the desperation for survival vanishes. Society shifts from a zero-sum game of scrambling for limited resources to a positive-sum game of continuous accumulation. As Mencius observed, "Those with a secure livelihood have a steady mind." Peace may be a moral achievement; but physical stability is what made it affordable. In England, even the fiercest political conflicts, from the baronial wars to the Civil War and the Glorious Revolution, were distributional struggles over power and revenue, not existential crises driven by famine or systemic collapse. Grassroots uprisings, though sometimes violent, occurred within a social order whose agricultural baseline remained intact; participants had livelihoods to return to once grievances were addressed or suppressed. By the standards of Eurasian continental history, where subsistence crises routinely escalated local unrest into dynastic overthrow and mass displacement, English political turbulence was structurally contained. The Thirty Years' War is the starkest illustration: its catastrophic toll derived less from combat than from the famine and epidemic it unleashed upon an already climate-stressed food supply, confirming that even the most devastating wars owed their systemic destructiveness not to combat as such but to their coupling with underlying food crises.

  In England, and to varying degrees across the Low Countries, northern France, and the American North, secure livelihoods lowered social friction and freed individuals from reliance on lineage-based security. They lived in nuclear families, standing in marked contrast to the kin-based extended families of highly volatile regions. This social structure allowed people to leave the land for industry, leading to lower birth rates and a more balanced population structure. This, in turn, promoted agricultural progress and paved the way for industrialization.

- **Easy Taxes**

  In highly volatile regions, tax collection becomes a constant struggle. With a tax base both volatile and difficult to assess, the state naturally suspects taxpayers of concealing wealth, while taxpayers naturally fear the state taking too much. This mutual distrust forces the state to intrude constantly upon private property to verify and seize resources, effectively blurring the line between taxation and confiscation.

  In England, however, under peaceful conditions, the government did not need intrusive and unpredictable tax collection. Internally, low social friction reduced the need for suppression; externally, island geography minimized the need for a standing army. The predictability of harvest made the tax base transparent. The government could collect a fixed revenue without intrusive monitoring, leaving property boundaries undisturbed.

Property rights protection may well owe to state benevolence; but it was the low transaction costs of taxation that kept it viable: when the fiscal base was transparent and collection straightforward, the state had no need to intrude upon private boundaries—intrusion itself carries costs of monitoring, enforcement, and resistance.[12]

- **Tolerable Administration of Justice**

  This point is the most misunderstood. The fundamental reason England could practice a loose and piecemeal common law system was that social friction was naturally low.

  In highly volatile regions, frequent disasters turn every contract into a dispute (e.g., tenants demanding rent abatement), generating a massive demand for judicial intervention and disaster relief. Beyond justice and relief, volatility also demands heavy investment in disaster-prevention infrastructure (dikes, granaries) and social order maintenance during crises. This creates a "Governance Paradox": it generates a massive demand for public services while also weakening the fiscal capacity to fund them. With a tax base both volatile and difficult to assess, the state lacks the fiscal resources to maintain a professional bureaucracy of sufficient scale, forcing it to rely on low-cost local proxies such as lineages or gentry to extend its reach. While this maintains basic order, it is a far cry from the efficient, uniform, direct administration imagined by modern observers. England, conversely, enjoyed a double blessing: low social friction reduced the cost of governance, while stable revenues ensured the capacity of governance. Justice could remain merely "tolerable", loosely structured, reactive, and locally administered, because the system was rarely pushed to the breaking point.

England enjoyed these conditions in their purest form. Across the Channel, the Low Countries and northern France shared much of the same stability, though with somewhat higher fiscal burdens imposed by continental defense. Stability made limited government viable; political contestation gave it form. England arrived at common law and parliamentary monarchy, the Netherlands at Roman-Dutch law and urban republicanism.[13] The foundation of social coordination was not uniquely English, but English conditions were uniquely favorable.

---

[12] Output volatility determines the enforceability of fixed claims wherever output is divided between a fixed claimant and a residual claimant. Sovereignty is the supreme ownership right over territory. The political community and the individual property holder share ownership: the community retains a collective claim on output (taxation), and the individual holds the residual (property rights). Property rights protection, then, is not a separate institutional category but a specific condition within this shared-ownership structure: it describes a state in which the boundary between collective claims and individual holdings remains stable. Whether this boundary holds firm is shaped by political will and institutional design, but fundamentally constrained by output volatility. In a low-volatility environment, the tax base is predictable and transparent. The collective claim can be fixed: the state can collect a fixed revenue without intrusive monitoring. The boundary between collective claims and individual holdings holds firm, and this stability is what we call property rights protection. In a high-volatility environment, the tax base becomes both volatile and difficult to assess. The state, unable to predict or verify revenues, must repeatedly intrude upon private holdings to extract resources. The boundary between collective claims and individual holdings becomes chronically unstable, and property rights protection becomes structurally unviable. This analysis illuminates the fiscal dimension of property rights, which is the dimension most directly shaped by output volatility. Within the boundaries that material conditions set, political agency, cultural inheritance, and historical contingency further shape the specific form of property rights institutions.

[13] The influential literature on European state formation attributes parliamentary constraints and property rights protection to interstate military competition. Yet the core cases are concentrated in the low-volatility zone this article identifies. The same material logic determines the type of Leviathan a society can sustain: Acemoglu and Robinson's typology accurately describes the phenotype of governance but leaves the genotype unidentified. See Appendix D for the full argument.

The economic essence of the English success was the presence of a stable "grain anchor." In economic terms, this material predictability structurally reduced the transaction costs of social coordination. A well-functioning market is, at its core, an information system. When the price of food is steady, the signals across the entire price system are clear and reliable. This clarity dramatically lowers the costs of information gathering, contract negotiation, and contract enforcement across all sectors.

This foundational anchoring is the invisible prerequisite for the rule of law: material predictability must precede legal certainty. "Better laws" may well be gifts of wisdom; but they gain efficacy only where the transaction costs of enforcing them are low enough by a stable environment to make enforcement viable. In a highly volatile environment where food prices are chronically unstable, the resulting chaos renders even the most sophisticated legal codes unenforceable.

5. Spontaneous Industrialization

Before the Industrial Revolution ignited, the Agricultural Revolution had already prepared the essential fuel: a massive food surplus. In a stable environment, landlords and farmers of all kinds, freeholders, tenant farmers, large and small, had incentives to invest in land improvement. The spread of the four-course rotation (wheat, turnip, barley, clover), which eliminated fallow periods and boosted yields, was one prominent result. However, this technical leap was not merely a farming improvement; it was a dividend of stability. The system required dedicating significant land to fodder crops (turnips, clover) for soil regeneration and livestock. In highly volatile monsoon regions, where every inch of land was needed to grow grain for human survival, dedicating land to fodder crops instead of grain was a luxury few could afford. Even where rotations existed (e.g., rice followed by green manure crops), they were aimed at maximizing caloric density, not growing different crops for the market. This specific constraint reflected a broader reality: the constant threat of famine forced a focus on immediate caloric extraction, blocking not just commercial crop rotation, but any form of market-oriented farming that required long-term stability.

While agrarian stability was the foundation, the geographic lottery endowed England with three additional natural advantages:

- **Geopolitical Security:** The English Channel served as a natural moat, insulating the island from the continental wars that periodically disrupted accumulation in Europe.
- **Low Transport Costs:** No place in England is more than 120 km from the sea. A dense network of rivers and canals connected industrial centers (like Manchester) with ports at extremely low cost, unifying the domestic market.
- **Energy Abundance:** Vast, shallow coal deposits were located conveniently near the coast (e.g., Newcastle), providing cheap, accessible fuel for the coming Steam Age.

At this point, a unique historical convergence occurred. A stable and abundant food supply secured the workforce; abundant coal provided cheap power; water transport ensured low-cost

circulation; and island geography protected the accumulation of wealth. When these factors came together, market uncertainty and production costs reached a historical low.

On this material foundation, a well-functioning market formed naturally. Capital was no longer confined to landholding or commerce. With the assurance of a stable market, wealth owners dared to tie up their liquid resources in fixed assets—factories and machines. Industrialization was less a top-down design than a natural emergence: once the material threshold had been crossed, stable markets and long-term industrial investment flourished.

England was the first to cross this threshold, but the same convergence of stable agriculture, cheap transport, and accessible energy replicated itself across northwestern Europe and the American North, making the rise of the West a broad-based phenomenon rather than an isolated English miracle.

This account focuses on the material preconditions. The intellectual and institutional innovations that also contributed to industrialization, from the scientific revolution to patent law, had origins of their own that this framework does not attempt to explain. What it does explain is why these innovations, which appeared in various forms across civilizations, could accumulate into a transformative system only where the material foundation was secure.

A clarification is in order. England's common law, parliamentary monarchy, and the constitutional settlement of the Glorious Revolution were of great importance to England's own history. But they do not possess universal significance. Northwestern European countries, the United States, Japan, and modern China have each industrialized fully under their own distinctive legal systems, state structures, and political processes. What these cases share is not a common institutional form but a common material foundation: stable and sufficient agricultural output. The institutions they share are not specific legal or political designs but the set of distributional arrangements that this common foundation sustains: enforceable fixed-cost contracts, secure residual claims, and predictable fiscal boundaries.

6. The Supersized England: The American North

The American North was another winner of the geographic lottery, an enhanced version of the English pattern.

New England and the Mid-Atlantic states (like Pennsylvania and New York) share a similar temperate, humid climate. Although situated in a continental zone, the region enjoys a rare degree of climatic stability. Thanks to the moderating influence of the Atlantic Ocean and, in the interior, the Great Lakes, rainfall is relatively reliable year after year. While seasonal amounts vary, the timing is steady and predictable, free from the wild fluctuations of the monsoon. Thus, the American North was blessed with agricultural stability. In states like Pennsylvania, known as the breadbasket of early America, abundant yields and predictable prices were the norm. This predictability was the foundation of its social order. Socially, while the prevalence of

independent freeholders originated from colonial land policies, their ability to stay independent was a dividend of agricultural stability.

Agricultural stability meant that a single family could be a self-sustaining economic unit without needing to surrender its autonomy for protection. This prevented the slide into debt bondage, the recurring cycle of debt and dispossession that weakened the Roman Republic and has plagued agrarian societies throughout history. Instead, this stability cultivated a resilient class of independent farmers who provided the stable social foundation and consumer market for the coming industrial age. This stands in stark contrast to the American South, where greater climatic variability compounded by reliance on volatile cash crops created chronic fragility. Rather than a resilient yeoman class, this environment produced a marginalized underclass of "poor whites"—smallholders on marginal lands who, lacking the reserves to buffer against crop failures and volatile market prices, were unable to sustain their independence. After the Civil War, this highly volatile environment led to the widespread adoption of sharecropping, trapping both former slaves and poor whites in systemic poverty for nearly a century.

The natural setup for transport and power was equally lucky. Excellent natural harbors like Boston, New York, and Philadelphia dotted the coast. Inland rivers like the Hudson were navigable deep into the interior. The opening of the Erie Canal in 1825 connected the Atlantic to the Great Lakes, slashing transport costs. While New England lacked coal, it had powerful rivers for hydropower; nearby Pennsylvania had vast coal reserves to fuel the later Steam Age.

Stable and abundant food supply, abundant energy, cheap transport, and isolation from European conflicts—the American North perfectly replicated England’s material foundation. With a vast hinterland, it became a supersized England. Here, a well-functioning market formed spontaneously, and industrialization unfolded as a natural progression.

IV. Colonial Plunder: A Boost, Not the Engine

While the conventional institutional perspective seeks the cause of Western success in human wisdom, an alternative and widely influential narrative seeks it in human sin, arguing that industrialization was fueled primarily by colonial plunder.

History proves that colonial plunder alone cannot make a modern nation strong; Spain is the most telling example. Comparing the failure of Spain with the success of England is not about excusing the sins of the past. It is a lesson in how nations actually grow: external wealth only works if a nation already has a stable foundation at home.

As the first empire on which the sun never sets, Spain plundered an astonishing amount of gold and silver from the Americas following Columbus's voyage of 1492. If colonial plunder were the engine of industrialization, Spain should have become the world’s ultimate industrial powerhouse. Instead, it became merely a conduit through which gold flowed to its rivals.

Spain's heartland, the arid central plateau and Mediterranean coast, suffered from high grain volatility. Without a stable food supply, it could not support the long-horizon commitments required for a robust industrial base. Therefore, the massive wealth from the Americas was not invested in domestic production. Instead, it was dissipated on imports, conspicuous consumption, and costly wars. Gold flowed through Spain like water, enriching the Netherlands, Britain, and France—countries that possessed the internal stability Spain lacked. Sitting on mountains of gold, Spain failed to industrialize in step with its rivals and eventually declined into a second-rate power. Conventional accounts attribute Spain's decline to Habsburg imperial overstretch, silver-driven inflation, or a resource curse that crowded out domestic industry; yet these were symptoms of a deeper incapacity: without domestic agricultural stability, no empire can convert external wealth into self-sustaining growth. Tellingly, the one Spanish region that did industrialize spontaneously, the Basque Country, is precisely the one that combines a maritime climate with low grain volatility, mirroring the English pattern. Yet like the Dujiangyan oasis, it was too small to overcome the volatility that held the rest of Spain back.

In contrast, England developed a prosperous domestic economy (textiles, shipbuilding) based on agricultural stability before it ever became a global plunderer. When England later turned to empire-building, every penny snatched from abroad could be invested into a thriving local industrial cycle: building factories, upgrading technology, and expanding trade. England became the global industrial center not just because it had the claws to snatch wealth, but because it had the “stomach” to digest and multiply it.

Colonial plunder was undoubtedly a huge boost for the West, but it was not the engine. The true engine was a convergence of geographic advantages: stable and productive agriculture, convenient water transport, and abundant energy. The regions surrounding the English Channel (England, Netherlands, Belgium, northern France) and the American North shared this winning combination. With this engine, the West acquired the capacity to plunder the globe.

To attribute the rise of the West merely to plunder is morally satisfying but logically flawed. This focus on external causes hides the true internal threshold of industrial civilization. The Global South, lacking this jackpot, cannot simply replicate the Western path. China faced the same constraint—and had to forge a different way forward.

V. Traditional China: The Stagnation Trap

What Chinese civilization faced was not just natural abundance, but a brutal challenge imposed by the continental East Asian monsoon climate. Here, boundaries and contracts could not hold against the recurring shocks of floods and droughts.

1. The Fragility of Specialization

An ancient song traditionally attributed to Emperor Shun captures the essence of the monsoon:

> The fragrance of the South Wind eases my people’s anger.

The timing of the South Wind increases my people’s wealth.

This song reveals the two sides of the monsoon. When the "south wind" (monsoon) arrives on time, it brings rain and heat, creating immense wealth. But the monsoon is fickle. Once it is untimely, abundance turns into disaster. For millennia, this uncertainty of living at the mercy of the monsoon has been a constant threat to survival. But it was more than that: it was also a barrier to specialization.

Take the salt industry as an example. If salt workers focused only on making salt and traded with farmers for food, everyone would benefit: specialization raises efficiency, and exchange allows both sides to gain. But in reality, this specialization was fragile.

For the salt worker, the monsoon created a constant threat: In lean years (drought or flood), grain harvests failed and prices skyrocketed. Those who had to buy grain spent most of their earnings just to survive. Farmers fared no better—a failed harvest meant no surplus to sell and no income to spare. Either way, demand for salt collapsed. The salt worker was trapped: grain was unaffordable, and salt found no buyers. To avoid starvation, he dared not specialize in salt; he had to grow some food himself as a lifeline.

Historically, the government tried to force specialization by mandating hereditary occupations (e.g., salt workers) and establishing large-scale state-controlled workshops. But this forced specialization was equally fragile. It relied on the state's ability to guarantee food rations. When the state's fiscal capacity was strained by famine or war, the lifeline of grain became unreliable. These workers had to make do with inadequate relief or simply fled. Whether forced by the state or attempted by the market, specialization without a guaranteed food supply was a castle built on sand.[14]

The same logic applied to the farmer. In bountiful years, grain prices crashed; in lean years, there was no surplus to sell. So he remained trapped in poverty. He couldn't focus solely on farming; he had to weave cloth or do odd jobs to make ends meet. This logic extended to all livelihoods. Workers dared not specialize; farmers dared not specialize. Fundamentally, specialization requires a stable exchange ratio; high volatility destroys it.

2. Why No Factories?

People often blame stagnation on overpopulation, but this is only a symptom. If this massive population could be absorbed by factories, that would be a dividend, not a curse. Why factories failed to absorb this workforce needs a more fundamental explanation.

---

[14] The state's attempt to impose specialization by administrative fiat reveals a second class of actor other than individual agents: one that can pool dispersed surpluses and invest across time horizons no individual agents can afford. But this capacity is bounded by the same volatility that necessitates it. High volatility erodes the fiscal base on which state action depends; when famine or war strains that base, the guaranteed grain supply on which forced specialization rests breaks down.

Take the traditional cotton textile industry as an example. Before the Opium War, China produced an estimated hundreds of millions of bolts of cloth annually. This seems like a huge industry. But this was not the result of mass production. It was produced by countless family workshops across the empire, most of it consumed within the household.

Why did large textile factories fail to emerge in this massive sector? The fundamental reason was high geographic volatility, which killed the possibility of expansion in scale and capital accumulation.

First, on the demand side, high market volatility made large-scale operations economically unviable. In an economy of "feast or famine," purchasing power fluctuated in lockstep with the harvest.

- In bountiful years, when food was cheap, families had money left over to buy cloth.
- In lean years, people couldn't even afford food. Demand for cloth would crash.

These recurring shocks meant that any factory owner who invested in machines and hired workers faced a severe threat: frequent demand slumps, triggered when spikes in grain prices drained purchasing power, made it impossible to sustain cash flow. He would eventually be buried by unsold inventory and fixed costs.

Second, on the supply side, capital could not be accumulated from the bottom up. Large-scale production requires sustained reinvestment. But in this environment, small producers spent most of their earnings buying expensive grain in lean years just to survive. The surplus created in bountiful years was eaten up by the deficit of lean years, preventing the accumulation of capital to build factories.

Finally, even those with capital (such as landlords or merchants) dared not invest in factories. Given the high uncertainty, they preferred land, short-term lending, or hoarding, which offered quicker returns or tangible security. Industrial capital was effectively blocked from forming.

The result was a structural impasse imposed by high volatility. As Adam Smith said, "the division of labor is limited by the extent of the market." In traditional China, the binding constraint on market extent was not geographic reach or population size, but high volatility. In such a market, the factory system was economically irrational for all parties: workers could not afford to leave the land, and capital avoided being tied up in factories. This explains why the workforce remained stuck in households and why the Industrious Revolution (the intensification of household labor) in China never led to an Industrial Revolution.[15]

---

[15] The micro-level mechanisms described here, agents diversifying away from specialization and capital avoiding high-fixed-cost investment, are well supported by the empirical literature on risk and agricultural behavior in developing economies. Rosenzweig and Binswanger (1993), using rainfall variability data from Indian villages, show that households in high-volatility environments systematically shift their asset portfolios toward lower-risk, lower-return investments, with the effect strongest among poorer households, a finding that directly parallels the capital behavior described above. Morduch (1995) distinguishes ex ante income smoothing (adjusting production decisions before shocks occur) from ex post consumption smoothing (borrowing or drawing down savings after shocks), and argues that the former has been underappreciated: poor households in risky environments sacrifice

3. The Self-Reinforcement of Involution

High geographic volatility, through its effect on grain output, intensified population pressure on land through two dynamics.

First: Tied to the Land. Because of frequent floods and droughts, farmers dared not leave the land. Otherwise, they might not be able to afford food in lean years. Therefore, no matter how much the population grew, everyone had to cling tightly to the land as a lifeline.

Second: The family safety net. Since life was hard and crowded, why have more children? Because in a highly volatile environment, the family was the primary safety net. The belief that "more children, more blessings" was not a superstition, but a rational survival strategy:

- **Labor and mutual aid:** More hands meant not only more help in the fields, but also a lifeline during disasters. When crops failed, the extended family pooled their food and labor to get everyone through the crisis.
- **Collective defense:** A large extended family was not just for mutual aid, but also a powerful deterrent. In a world where law and order were weakened by the chaos of nature, such numbers discouraged bandits, thieves, or even greedy neighbors from preying on your household. In a highly volatile environment, there was truly safety in numbers.

These two dynamics created a trap far more complex than simple poverty: “Involution.” Intensive labor input with diminishing marginal returns: as more hands worked the land, output per person declined.[16] Farmers farmed and wove at the same time. The whole family worked long hours with high intensity, yet they remained extremely poor. This was not because they were lazy or stupid, but because specialization was economically unviable in such a highly volatile environment. Worse still, involution left the system dangerously fragile: a dense population surviving on minimal surplus, with no buffer against consecutive bad harvests. Individual

---

expected income to reduce volatility, choosing safer but less profitable crops and diversifying into non-farm activities. This is precisely the 'rational diversification under endemic uncertainty' that this framework identifies as the micro-foundation of the involution trap. Townsend (1994) tests full risk-sharing in Indian villages and finds that village-level insurance is substantial but incomplete: individual income shocks still affect consumption. This confirms the framework's treatment of kinship and village networks as meaningful but insufficient buffers, palliatives operating within the constraint rather than solutions that remove it.

[16] Population density and population pressure are distinct concepts that must not be conflated. Density is a static measure: the number of people per unit of land. Pressure is a dynamic, functional condition: the competitive squeeze of population against the carrying capacity of the land, manifested in declining per capita resources, the inability of labor to leave agriculture, and the progressive narrowing of the margin of survival. The level of output determines the upper bound of density a region can sustain; the volatility decides the fate of that density: under stability it is channeled into specialization and industrial development; under recurrent shocks it is locked into involution and environmental degradation. The Low Countries are dispositive. The Netherlands and Belgium sustained some of the highest population densities in early modern Europe, yet neither exhibited the involution trap characteristic of the Yangtze Delta. The reason is clear: low output volatility meant that dense populations could specialize, leave the land, and enter industry. Density translated into market extent and deepening division of labor rather than into the survival pressure that blocks both. Conversely, in the Yangtze Delta, comparably high density was converted into pressure precisely because high volatility tied labor to the land, incentivized large families as self-insurance, and prevented the factory system from absorbing surplus labor. If density itself were the binding constraint, the Low Countries should have involuted; they did not. What distinguishes involution from industrialization at high density is not the number of people but the volatility of the output that sustains them.

strategies for risk-minimization aggregated into a systemic fragility that none had intended.[17] As R.H. Tawney observed of rural China: "There are districts in which the position of the rural population is that of a man standing permanently up to the neck in water, so that even a ripple is sufficient to drown him."

While this tendency was widespread across highly volatile environments, it found its most extreme expression in rice-growing regions like the Yangtze Delta. Unlike wheat, wet rice cultivation acts as a "labor sponge"—it allows for an exceptionally high degree of labor intensification (e.g., hydraulic management, heavy fertilization) to squeeze out marginal gains.

This immense labor input did, in fact, mitigate the volatility of yields and allowed for significant capital accumulation. The region became wealthy, and a commercial class emerged. But this wealth did not arise from large-scale farming or factory production. It accumulated through traditional channels: landlords collecting rent, merchants profiting from long-distance trade in silk and cotton cloth, and moneylenders earning from short-term credit.[18]

---

[17] In pre-modern societies, human activity can amplify geographic volatility but does not generate it at the systemic level. Where pre-modern technology could not suppress the volatility that nature imposed, human action adapted to it rather than overcoming it. The climatic baseline thus sets not only the initial level of volatility but also the direction of the human behavioral response, producing a positive feedback loop whose sign is set by the environment itself. In a high-volatility environment such as China, climatic volatility made large families a survival necessity, intensifying population pressure; this pressure in turn drove behaviors that amplified volatility (clearing hillsides for cropland, reclaiming lakes), which further reinforced the incentive for large families. The feedback is self-amplifying: volatility begets the very human responses that deepen it. In a low-volatility environment such as England, the same feedback operates in reverse. Geographic stability supported nuclear family structures and moderated population growth, reducing the pressure on land and forests. England underwent comparably thorough deforestation, with forest cover falling to roughly 8 to 10 percent by 1700 and reaching about 5 percent by the late nineteenth century (Rackham, 2006), yet this did not produce systemic flood-drought cycles. Two factors account for the difference. First, the maritime climate and moderate terrain kept rainfall stable regardless of land cover. Second, the climatic baseline shaped land use itself: England's reliable rainfall made permanent pasture highly profitable, so much of the cleared land became grassland rather than ploughed fields. Continuous grass cover retains the hydrological function of forest, absorbing rainfall and minimizing runoff. In China, the same pressure to clear land was directed almost entirely toward cropland, often on steep hillsides, where periodic exposure of bare soil during tillage accelerated erosion, silted rivers, and intensified flooding. Thus even the mode of deforestation diverged along lines set by the climate. The cross-case comparison confirms causal primacy at two levels. First, deforestation of comparable scale produced divergent hydrological outcomes, establishing that the climatic baseline, not human activity, is the systemic determinant. Second, the low-volatility environment suppressed the behavioral dynamics (high fertility, extreme land-use pressure, erosion-prone cropping on marginal land) that would have tested its resilience, while the high-volatility environment actively generated them. The climate baseline shapes not merely the starting point but the boundaries within which the entire system evolves.

[18] The apparent commercial integration of traditional China requires qualification. Much of the long-distance trade that gave the empire its appearance of a vast unified market was not autonomous commerce operating under price signals, but was substantially embedded in state fiscal and administrative functions. The most prominent examples: (1) The salt trade, one of the empire's largest commercial sectors, operated under state monopoly. Merchants purchased government-issued salt certificates and competed within a licensing framework whose primary purpose was fiscal revenue extraction. The great merchant houses of Yangzhou—among the wealthiest in the eighteenth-century world—derived their fortunes principally from state-granted monopoly privileges. (2) The rise of the two most powerful merchant groups, the Shanxi merchants and the Huizhou merchants, was significantly catalyzed by the early Ming *kaizhong* system, under which merchants delivered grain to frontier military garrisons in exchange for salt certificates. Their commercial networks expanded outward from this fiscal-military nexus. (3) Even the cotton cloth trade, the largest sector by volume, was partly shaped by state fiscal imperatives: in earlier periods, households in designated regions owed tax payments in cloth, and the state redistributed it across the empire, though this administrative component diminished as the tax system progressively monetized. These examples suggest that the "extent of the market" in traditional China, measured by the sheer volume of goods in motion, overstates the degree of autonomous market coordination through price signals. A significant share of what appeared to be commercial activity was organized by, or dependent upon, the

Given that intensive labor had already mitigated yield volatility in the Yangtze Delta, and that wealth had accumulated, why did this capital never flow into factories? The answer lies in involution, the survival strategy itself, which came at a crushing economic cost: the devaluation of labor. Chronic labor surplus, aggravated by agricultural seasonality, pushed the price of labor to the subsistence floor. This cheap labor neutralized the power of the accumulated capital to trigger industrialization:

- **The Fragility of Demand (Demand Side):** Low labor value meant that despite the region's aggregate wealth, the vast majority of households lived with minimal financial buffer. They remained extremely sensitive to grain price volatility. A slight rise in food prices would wipe out their purchasing power for industrial goods, making the mass market too unstable for factory investments.
- **The Capital Diversion (Supply Side):** As we will explore next, the abundance of cheap labor destroyed the economic rationale for machinery investment. Even though capital was available, it did not flow into machinery, because capital itself was expensive while labor was cheap.

Thus, the economy achieved "Growth without Development": capital accumulated, but the very conditions that enabled it blocked its industrial outlet.

4. Rational Regression

It is generally recognized that in traditional China, the smallholder economy blocked industrial progress. But there were exceptions—moments when advanced machinery did appear. These leaps were typically driven by the state or local elites, who possessed the capacity to concentrate capital and mobilize resources for high-level engineering. As one of these “technological peaks,” the water-powered Grand Spinning Wheel, documented during the Yuan Dynasty, represented the technological frontier of the agrarian era. Used for spinning hemp, it had 32 spindles and was driven by a massive water wheel. Its complexity was notable.

But the seeds of stagnation were sown in this very prosperity. By the Ming Dynasty, two things changed: cotton replaced hemp, and driven by the involution trap, population pressure surged to a saturation point. As we explained, the involution trap forced people to stay on the land and

---

administrative apparatus of the state.

The regional origins of the scholarly evidence suggest a parallel overstatement at the level of local market studies. The most influential account of China's market system, Skinner's model of rural market structure (1964 – 65), was built on fieldwork conducted on the Chengdu Plain, whose core agricultural zone falls within the Dujiangyan irrigation system that this article identifies as an engineered low-volatility oasis. The California School's broader claims of market development comparable to England's (Pomeranz, 2000; Wong, 1997) drew primarily from the Yangtze Delta, where massive labor input had partially buffered volatility. Neither tradition engaged primarily with the North China Plain, where monsoon volatility was least buffered. When Philip Huang (1985) did examine the North China Plain directly, he found a qualitatively different kind of market exchange: commercialization driven by fiscal and debt pressures rather than voluntary specialization, combined with high self-sufficiency and acute household fragility. The scholarship that established the image of a highly commercialized traditional China thus drew its evidence almost entirely from regions where volatility had been most effectively suppressed, by ancient engineering in Sichuan and by labor intensification in the Yangtze Delta.

reproduce, driving the market price of labor to the subsistence floor. The same volatility that depressed labor costs also drove up the price of capital: with lenders demanding high returns on short-term loans to hedge against recurrent shocks, the financing cost of machinery was prohibitive. Since human labor was cheaper than employing a complex machine, labor-saving technology became unviable.[19] Cheap labor and expensive capital together locked out mechanization. Although the Grand Spinning Wheel was advanced, it lost its economic rationale. The chain of technological improvement broke, and it failed to evolve into a version for cotton.

The fate of the Grand Spinning Wheel is a microcosm of countless inventions in traditional China. Like meteors, they flashed across the sky and vanished from history, swallowed by the involution trap. The disappearance of such machinery was not an accident; it reflected the same structural impasse that would block industrialization for centuries. Even where intense labor input achieved relative stability in grain yields and allowed for capital accumulation (as in the Yangtze Delta), the combination of cheap labor and expensive capital killed the incentive to mechanize, while highly volatile markets killed the factories that could house machines. Consequently, this capital could not find an industrial outlet. Without the economic conditions to make technological upgrading pay off, even the best machines cannot ignite an Industrial Revolution.

VI. Contemporary China: A Narrowly Won Big Push

To understand the first thirty years of the People's Republic, we must answer one core question: What was the imperative of the revolution?

From the late Qing to the Republican era, traditional agriculture was buckling under population pressure. External shocks from the West broke the old order. Famine, warlords, and invasion pushed Chinese society to the brink of chaos. In this environment of political disorder compounded by geographic volatility, price signals were drowned out by survival panic and speculative hoarding, and market mechanisms could not sustain the coordination that industrialization demands.

The revolution, therefore, was not merely an ideological choice, but a necessity for survival. Its success lay not just in seizing power, but in reestablishing a viable order out of chaos. But lasting order in a monsoon environment demanded more than political consolidation; it required securing the material foundation that political chaos alone had not destroyed but geographic volatility had always undermined. The century of humiliation had demonstrated that an agrarian nation could neither withstand industrial powers nor sustain internal order under recurrent natural shocks; industrialization was not a choice but a condition of survival. To industrialize, China first had to overcome its age-old vulnerability to geographic volatility. This was a massive undertaking of engineering: a total reconstruction of the nation’s landscape.

---

[19] This account complements Robert Allen's influential 'high-wage hypothesis' (Allen, 2009), which attributes England's mechanization to high wages relative to cheap energy. Within the endowments-volatility framework, England's comparatively high wages were themselves a product of geographic stability: low volatility supported nuclear family structures and moderated population pressure, keeping the labor force relatively scarce and wages high. Allen's factor-price incentive for mechanization was thus downstream of the volatility conditions that this article identifies as the deeper determinant; cheap coal widened the gap further but did not create it.

From this perspective, the history of the first thirty years takes on a new meaning. It was not a “detour violating comparative advantage” (i.e., prioritizing capital-intensive heavy industry despite abundant labor and scarce capital), but a Big Push that overcame the binding constraint. For late-developing nations facing systemic underdevelopment, a Big Push must weather the fundamental challenge of grain volatility and ultimately overcome it through the very power generated by industrialization, or the Big Push will fail to deliver self-sustaining growth. This is inherently a perilous path: the nation must extract surplus from an already fragile agricultural base to fund industrialization, while that very base remains vulnerable to the shocks industrialization has not yet tamed. The margin for error is razor-thin, and the human cost of missteps can be devastating.

Yet the risk of inaction was no less real. Without industrialization, China would remain an agrarian nation militarily defenseless against industrial powers, as the century of humiliation had proved, while the recurrent famines that had devastated the population throughout Chinese history would continue unabated. China faced no risk-free option; the Big Push was a calculated gamble not against safety, but against a baseline of perpetual vulnerability.

1. The Planned System: A Buffer against Volatility

At the founding of the People's Republic, China was essentially still a pre-industrial agrarian society. Grain production was volatile, hovering precariously around survival levels. It was in this desperate context that China adopted the planned system.

Critics often blame the planned system for the shortage economy. Shortages were undeniable, but this criticism confuses cause with effect: it was not the plan that caused the shortage; it was the shortage, chronic scarcity and volatility, that necessitated the plan.

Indeed, China had suffered from shortages long before the planned system existed. In the past, shortage meant famine; industrial capacity was equally absent. As Mao Zedong famously lamented: "What can we make now? We can make tables and chairs, teacups and teapots, we can grow grain, grind it into flour, and make paper, but we can't make a single car, a single plane, a single tank, or a single tractor."

Yet, industrialization requires capital—and capital must rest on a currency that holds real value. But what gives money its value? The value of money is derived not from gold, but from the command over wage goods—essential commodities like food and clothing that people need to survive. As Chen Yun, the architect of China's early economy, insisted, the amount of currency that could be issued was determined by the supply of physical goods, above all, grain and cotton cloth. **At its foundation, money holds value only when it can, at minimum, command grain.**[20]

---

[20] This is not a primitive monetary theory but a precise judgment about the hierarchical structure of currency anchoring. Monetary value rests on a material foundation: the stable supply of food and energy that ensures currency can command the means of survival and production. The principle is equally visible in energy: the US dollar sustains its role as the world's reserve currency in no small part because it can command oil. Above this foundation sit the sophisticated instruments of monetary management: central bank policy, credit systems, sovereign debt markets. In a mature industrial economy where food and energy supplies are reliably secure, the

In a highly volatile environment, limited grain surplus was precarious; uncoordinated market exchange would be chronically vulnerable to hoarding and speculative price spirals, preventing the price stability that industrialization requires from taking root. By monopolizing grain procurement and distribution through administrative channels, the state removed grain from speculative market exchange and fixed its effective exchange rate against the currency.

Thus, here lies the essence of the planned system: the state used its limited grain surplus as an anchor for the currency, engineering an insulated industrial sphere as a substitute for the market. Within this safe haven, heavy industry and research sectors could accumulate capital and advance technology, isolated from external turbulence.

The Austrian critique of central planning, most forcefully articulated by Hayek, holds that planned prices are mere accounting units that destroy the information system on which efficient allocation depends. In a mature market economy this critique carries great force. But in the environment the planned system actually inherited, price signals were already too obscured by volatility to guide full specialization. Without administrative shielding, production would not have advanced to a higher level of market coordination; it would have remained trapped in the partial specialization of the pre-modern era. The efficiency losses of central planning were real, but they must be weighed against this actual alternative, not against an idealized market that did not exist.

The planned system served as a buffer against high volatility, sustaining the nascent industrial base. It represented a real advance in the division of labor over pre-modern household self-sufficiency: workers in state factories specialized in single tasks, and the industrial system achieved a degree of coordination impossible under autarky. But its efficiency was constrained by the planner's limited access to dispersed information, falling well short of what a fully functioning market could achieve.

2. The Silent Agricultural Revolution

In the first thirty years, China completed a deep transformation of nature with one core goal: neutralizing survival risks.

---

foundation recedes from view, and economists naturally focus on the upper-layer mechanisms that appear to govern monetary value. But this visibility is itself a dividend of stability: the foundation has not been transcended, only obscured. Modern central banks institutionalize this obscuring: core inflation indices strip out food and energy prices as 'noise,' focusing monetary policy on the residual. This is analytically valid only so long as food and energy supplies are stable enough to make their price fluctuations transitory. Each time that premise has broken down, from the 1970s oil crises to the 2007-08 food crisis to the 2021-23 inflation episode, the stripped-out 'noise' has overwhelmed the upper-layer framework, forcing emergency policy reversals. The practice of excluding food and energy from the analytical baseline is itself a dividend of the material stability that this article identifies as the foundation of economic order. Chen Yun's insistence that currency issuance be tethered to the physical supply of grain and cloth reflected his recognition that, in the China of the 1950s, the upper layers of monetary architecture could not yet be built. The only available starting point was the most fundamental one.

The Big Push: centered on 156 key industrial projects backed by the Soviet Union, the Big Push built a complete industrial backbone. But more importantly, it laid the foundation for the agricultural revolution.

The overlooked battleground: the most critical yet underappreciated achievement was the systematic engineering in agriculture. To achieve stable yields under the monsoon climate, a thorough re-engineering of the physical landscape was necessary. By focusing on four key needs (water, fertilizer, pesticides, and better seeds), China achieved a silent agricultural revolution:

- **Irrigation and Flood Control:** upon the founding of the People's Republic, a massive struggle to tame rivers began, from the Huai River campaign to the countless reservoirs across the country. By the late 1970s, China had built approximately 80,000 reservoirs (National Bureau of Statistics of China, various years), irrigating in drought and draining in flood.
- **Fertilizer:** starting from scratch in the 1950s, China invested heavily in fertilizer plants, importing 13 massive complexes in the 1970s. By 1980, usage reached 12.7 million tons, and by 1990, it exceeded 25 million tons (National Bureau of Statistics of China, various years). This surge in the early 1980s meant agriculture finally broke its natural limits: utilizing industrial chemistry to turn fossil fuels into nitrogen for the soil. This was the critical tipping point where per capita output shifted from stable to soaring.
- **Pesticides:** a basic pesticide industry was established to curb pest outbreaks, helping reduce crop losses.
- **Seeds:** breakthroughs in crop breeding (including hybrid rice, improved wheat varieties, and hybrid maize) significantly increased yields while making crops more resistant to pests and unpredictable weather.

The impact of this revolution was transformative. In the first thirty years, while the population nearly doubled (from 540 million to nearly 1 billion), per capita grain output was broadly maintained around 300 kg (National Bureau of Statistics of China, various years). Maintaining per capita output despite a population explosion was a significant outcome. By the early 1980s, as these investments matured, grain yields rose sharply. Per capita output climbed steadily, breaking the Malthusian ceiling.[21]

This aggregate outcome was neither smooth nor costless. The twin pressures of military threat and food scarcity generated sustained political strain, which at critical moments erupted into catastrophic episodes. This was not an inevitable cost of the Big Push but a policy disaster born of political failure. Just as the planned system inherited rather than created the dysfunction of price signals, these catastrophic events cannot be attributed to centralization alone. In the absence of a centralized state, the same volatile environment had produced its own catastrophes for millennia: famine, banditry, warlordism, and dynastic collapse. In a high-volatility environment,

[21] In the terminology of agricultural economics, China's package of modern inputs did not merely compress variance; it reshaped the entire distribution of output. Under the pre-modern monsoon regime, agriculture was structurally left-skewed: the upper bound was capped by land and technology, while the lower bound could collapse toward zero under drought or flood. The systematic combination of irrigation, fertilizer, and improved seeds simultaneously raised the mean, compressed the variance, and effectively eliminated the catastrophic left tail, shifting the distribution toward symmetry. This is the statistical content of what this article terms 'engineered de-risking.

information is already obscured by recurrent shocks. Here, centralization was an advance, organizing survival where none had been organized before, but as an adaptation to this environment, it cannot render itself immune to that obscuring. When the margin between sufficiency and starvation is razor-thin, the severity of the obscuring is amplified, and the consequences become catastrophic when a single miscalculation propagates across an economy with no buffer to absorb it. The parallel with England is instructive. England arrived at its governance equilibrium only through costly political struggle: traditional royal prerogatives were progressively curtailed, and when the Stuarts resisted, the nation paid the price of civil war, confirming that in a stable environment, concentrated authority was more costly than it was worth. China's most severe governance crisis exacted a comparable price: under sustained political strain, the state pushed its command over the economy and society beyond the breaking point, with catastrophic consequences, and the surviving consensus shaped a more pragmatic approach. In both cases, equilibrium was not designed in advance but discovered through painful experience. The human cost of both episodes was enormous. But the margin of error determined the form of the catastrophe: England's stable agricultural baseline meant that political miscalculation manifested as war deaths; China's razor-thin surplus meant that political miscalculation manifested above all as hunger, though at times also as violent upheaval.

The conventional narrative credits the Household Responsibility System, the return to family farming in the early 1980s, as the decisive cause of the agricultural boom. This view rightly recognizes the role of incentives but understates the material preconditions that made them effective. China had practiced household farming for millennia; it had never, by itself, solved the problem. In a highly volatile environment, no incentive mechanism, however well-designed, can overcome the systemic risk that destroys surplus as fast as it is created. What changed by 1980 was not the incentive structure alone, but the physical environment in which it operated. Reservoirs tamed floods and droughts; fertilizer broke natural yield ceilings; pesticides reduced crop losses; improved seeds raised yields and resistance to weather variability. It was this engineered stability that made household incentives productive for the first time in Chinese history.

In contractual terms, the Household Responsibility System was structurally a fixed-rent arrangement: the household owed a fixed quota to the collective and the state, and retained all surplus beyond it. This is precisely the logic of the English tenant farmer paying a fixed rent and claiming the residual. China had known fixed rent for centuries. The *iron-rent* of the Yangtze Delta was its most recent incarnation. But as we showed earlier, in a highly volatile environment, fixed rent remained nominal across most of the country: disasters forced repeated abatement, collapsing it back into de facto sharecropping. Genuine rigidity existed only in rare engineered oases like the Dujiangyan irrigation zone, too small to alter the national trajectory. What made the post-1980 fixed quota genuinely rigid was the same factor that made English fixed rent rigid: low output volatility. Engineering had suppressed the volatility to the point where the quota could be reliably met year after year, and the household, now a genuine residual claimant, had every incentive to maximize output. The Dujiangyan logic, once confined to a 2,000-square-kilometer oasis, had finally been replicated at a continental scale.

The resulting grain abundance served a function far beyond household wealth: it stabilized the national food supply, anchored the currency, and released hundreds of millions of hands from the land into industry, setting the stage for the virtuous cycle that followed. The sequencing matters: first, the collective system built the material foundation; then, the restoration of household incentives released the energy that the foundation could now support. This was the deepest institutional change in China's history—a transformation not of political form, but of the baseline institution itself, made possible only by the reconstruction of the material foundation upon which all institutions rest.

Looking back, the Big Push was driven by twin imperatives, military survival and agricultural stability, pursued simultaneously under extreme resource scarcity. The convergence was structurally inherent: the heavy industries built for national defense were precisely the industries needed to engineer agricultural stability. Nuclear deterrence further eased the tension between the twin imperatives: by securing survival against external threat at far lower ongoing cost than conventional forces, it moderated the military claim on the same industrial capacity that agricultural de-risking required. The framework advanced here does not claim that the architects of the Big Push foresaw this outcome. It claims that the material logic connecting heavy industry to agricultural de-risking explains why the Big Push ultimately produced self-sustaining growth, a structural outcome that neither the planners' original blueprint nor the standard comparative-advantage critique can fully account for. The planners set out to secure survival; that the material foundation they built would ultimately render their own planning system obsolete was an outcome none of them intended.

VII. Contemporary China: Self-sustaining Growth

The planned system had served as a safe haven, nurturing infant industries in an insulated environment. But its capacity was inherently limited: with a weak currency anchor rooted in a limited grain supply, the system could only sustain an administratively coordinated division of labor, more specialized than pre-modern autarky but constrained by the planner's limited access to dispersed information. Once the silent agricultural revolution crossed the material threshold, a well-functioning market could finally take root and thrive. The full-scale industrialization of the last forty years was no longer driven by the state's Big Push, but became self-sustaining once the stable foundation had been built. The altered material conditions made the shift toward market coordination the natural direction; the process, as with all institutional change, involved real political contestation and carried no guarantee of a smooth path.[22]

[22] The transition from the planned system to the market system was not an external correction imposed upon a failed experiment. The planned system arose as a rational adaptation to high volatility, mobilized resources to engineer the stability that nature had denied, and in doing so eliminated the conditions that had made it necessary. It negated itself not through failure but through success. See Appendix E for the full argument.

1. The Virtuous Cycle Begins

The foundations laid during the first thirty years finally enabled a breakthrough in the early 1980s.

First, the silent agricultural revolution mitigated the effects of geographic volatility. A stable food supply provided a solid anchor for the currency. With food finally abundant and accessible, the rationing system and strict controls on population movement loosened, and the real costs of labor and commodities became predictable. On one hand, lured by the prospect of industrial profits and tax revenue, investors and local governments mobilized to build factories. On the other hand, hundreds of millions of farmers could finally leave the land and enter factories. This ultimately ignited the explosive industrialization spearheaded by Township and Village Enterprises (TVEs) in the 1980s.

Second, the state slashed the factor costs of production. Railways and roads expanded steadily, supporting large-scale logistics. A robust power grid gradually covered the country, providing more reliable energy than anywhere else in the developing world. This extensive transport and power network allowed the production system to operate at extremely low costs, integrating scattered regional markets into a unified national market.

At this point, a convergence comparable to England's occurred, but engineered rather than providential. A stable food supply, military security, cheap transport, and reliable energy came together. Thus, we witnessed one of the most rapid and extensive expansion of the division of labor in human history. Hundreds of millions of farmers entered factories to specialize in a single task. Spindles spinning in textile mills, hands assembling in electronics factories, everyone did only one thing, and productivity skyrocketed.

By unleashing the Smithian logic of specialization across its vast workforce, China became the “world's factory”. Opening up to foreign investment, technology transfer, and ultimately WTO accession in 2001 provided powerful multiplier effects, dramatically accelerating the process that the material foundation had made possible. This transformation went beyond cheap labor; it was the deepening of specialization itself—the true source of China's manufacturing power. Compared to the Global South, where industrial systems remain fragmented, China possesses highly integrated supply chains. Compared to the West, where aging infrastructure often constrain efficiency, China possesses substantial scale and structural cost advantages. Ultimately, this all-round strength allowed Chinese manufacturing to reach historically unprecedented levels of efficiency and scale.

More importantly, China has not stopped there. Building on its integrated supply chains, China is spearheading the New Energy Revolution, reinforcing the material foundation of its market.

If the Agricultural Revolution solved the problem of food for humans, the New Energy Revolution is solving the “problem of food for machines”. Breakthroughs in nuclear, solar and wind power, as well as battery storage, mean energy will become cheaper, more stable, and more ubiquitous

than ever before. This will further reduce production costs and iron out the risks of global energy volatility. In this sense, the New Energy Revolution is not merely an industrial advance but an enhancement of the engineered invisible hand itself.

This is a reinforcing feedback loop: a well-functioning market enables full industrialization, which generates new energy technology, which in turn reinforces the physical foundation for the market. This cycle pushes China toward the next frontier of the industrial revolution.

2. Engineering a Well-functioning Market

In analyzing the China model, Professor Justin Yifu Lin proposed the twin concepts of "facilitating state" and "effective market" (which we term "well-functioning market"). He argues that the government should provide infrastructure, coordinate investment, and facilitate industrial upgrading, while the market allocates resources. This is a profound observation. However, integrating this observation with history reveals a deeper causality.

For late-developing nations, a well-functioning market does not emerge spontaneously. Nor does it grow automatically just by hanging up a free market sign, privatizing assets, and freeing prices. A well-functioning market requires a solid anchor for the currency. This anchor rests on an abundant, stable supply of food and energy, achievable only through systemic engineering.

Therefore, the fundamental logic of the China model is: t**he state engineered the material conditions for a well-functioning market.**

Professor Lin also views the heavy industry drive of the first thirty years as a “detour”—a violation of comparative advantage. In this context, comparative advantage means that China, with abundant labor and scarce capital, should have developed light industry first, rather than jumping into capital-intensive heavy industry. Lin argues that this detour led to inefficiencies and shortages. But if we shift our lens from market-based resource allocation to the physical prerequisites that must be met before any market can function, a different rationale emerges.

Why did China insist on defying comparative advantage to develop heavy industry through self-reliance rather than pursuing the trade-based light-industry path that the comparative-advantage critique prescribes?

The Cold War blockade closed Western markets, making the import of advanced industrial goods and agricultural inputs impossible. But even absent that blockade, the trade-based light-industry path faced three insurmountable obstacles.

First, the labor advantage it presupposes was physically out of reach. A large population does not automatically translate into an available industrial workforce; without a stable food surplus, the massive population was tied to the land, forced to grow their own food just to survive. Even locally organized production, which does not require labor migration, never consolidated into the market-driven factory system, precisely because demand volatility destroyed the stable market that factory production requires.

Second, capital scarcity, which the comparative-advantage critique identifies as the binding constraint, was itself a product of food insecurity: volatile output repeatedly destroyed the surplus from which capital might have accumulated. The deeper constraint was food, not capital. Light industry's comparative advantage lies in labor intensity, yet in a food-scarce environment, every additional worker drawn from agriculture reduces grain output while adding a mouth to feed, aggravating food scarcity, and food scarcity is the constraint that overrides all others. Heavy industry, being capital-intensive, requires fewer workers per unit of output and therefore places less strain on the food supply.

Third, agricultural de-risking requires a permanent, continuously replenished supply of industrial inputs such as fertilizer, construction materials, and machinery, consumed and replenished year after year on a scale proportional to the population and territory to be secured. Military security imposes a parallel demand. Only heavy industry can produce these inputs. A nation may choose to source them externally, but it must then accept the vulnerability that comes with dependence on a supply line beyond its control. Even Soviet aid, the most substantial China ever received, encompassed 156 key projects, yet was abruptly withdrawn in 1960, underscoring the inherent fragility of external dependence. Importing industrial goods is no substitute for domestic capacity: equipment requires a domestic base to absorb and maintain; consumable inputs such as fertilizer and construction materials require continuous replenishment that no external supply line can permanently guarantee.

The East Asian economies often cited as counterexamples do not withstand scrutiny. Japan, already an industrialized power before its wartime defeat, was rebuilt under American occupation and remained within the American security umbrella. South Korea and Taiwan, far smaller, were even more thoroughly embedded. Their food security was underwritten by American grain shipments, their military security was guaranteed by American alliances, and their export revenues were stabilized by preferential access to American markets. In effect, these economies merged into America's low-volatility system, replicating at the geopolitical level the same pattern by which high-volatility regions within a single nation were pulled into industrialization by their low-volatility cores. The United States served as an external stabilizer for them much as England's maritime climate served as a natural stabilizer for its farmers: one was geopolitically provided, the other geographically given, but both performed the same function of suppressing the volatility that would otherwise have blocked industrialization. Their compact scale made this fusion feasible: the volume of grain, energy, and security commitments required was modest relative to American capacity. A continental-scale nation cannot be absorbed into another power's low-volatility system; the sheer volume of food, energy, and security commitments required would overwhelm any external guarantor. For a nation of half a billion people facing Cold War isolation, no such external stabilizer existed. China had to engineer its own stability from within, and only heavy industry could provide the means.[23]

---

[23] The East Asian economies often cited as counterexamples do not constitute relevant parallels. Japan was already an industrialized power before its wartime defeat and was rebuilt under American occupation. South Korea and Taiwan, far smaller, were even more thoroughly embedded in America's low-volatility system through security guarantees, grain shipments, and preferential market access. More broadly, no middle path between self-reliant heavy industry and full trade dependence has succeeded at continental scale. See Appendix F for the full assessment.

In short, the survival pressures China faced were twofold, military and agricultural, and only heavy industry could address both. Self-reliant heavy industry, however daunting the undertaking, was not an ideological preference but a physical necessity.

The logic thus comes full circle. To release labor from the land, agriculture had to achieve high and stable yields; to achieve that, China had to use heavy industry to engineer away the material constraints:

- Without steel and cement, there would be no dams or irrigation networks.
- Without an advanced machine-building industry, it would be impossible to manufacture the high-pressure equipment required for synthetic ammonia (the key to fertilizer).
- Without the chemical industry, there would be no pesticides.
- Without the power industry, there would be no rural electric grid to power the vast network of electric irrigation and drainage stations.

It was this industrial network that first secured the survival floor against droughts and floods, and then progressively drove yields beyond natural limits.

Therefore, the heavy industry strategy, while it appeared to violate comparative advantage through the lens of static efficiency, was actually an inevitable choice grounded in physical constraints. By channeling industrial capacity into agriculture, the state systematically neutralized the effect of geographic volatility and stabilized the price system. Once this material threshold was crossed, the comparative advantage in labor, which had existed only on paper, finally became an economic reality.

This is the core logic of the China model: while a well-functioning market may coordinate economic activity, it cannot bring itself into being. In a highly volatile environment, the state cannot just “facilitate” the market; it must engineer the material stability that allows a well-functioning market to exist in the first place. First, the state reshapes the material foundation and tames volatility. Then, and only then, can the market mechanism take root and thrive.

To truly understand the China model is to understand why some have argued that Deng Xiaoping should not be called the "chief architect" of reform and opening up, but rather the "chief approver": the most effective institutional reforms were not designed from above, but grew from below once the material foundation was in place.

VIII. Governance: Two Modes Compared

The difference between Chinese and American governance is often summarized as engineer governance versus lawyer governance. Yet, this distinction runs deeper than style. It traces back to the two origins introduced earlier: the law from Sinai, where order was received as a divine inheritance to be preserved, and the engineering from Tu, where order was physically constructed out of nature's chaos. This divide between preservation and transformation defines

the state's primary mission. In a stable environment, the state functions primarily as a referee of established rights, preserving a social order that nature has already made secure. However, in a highly volatile environment, the state's first mandate is to mobilize the people to engineer the very stability that nature denied. Here, the priority is not settling disputes, but actively creating the material conditions for survival.

To be sure, the Sinai narrative is a cultural archetype, not a causal origin. The actual Western model of preservation-oriented governance stems from the English experience. In that providential oasis, grain output was stable, society was peaceful, and external threats were minimal. The government did not need to battle highly volatile harvests, civil unrest, or invading armies. In this context, upholding established rights was all that was needed.

Thus, the West developed the ethic of procedural justice. Its core is preservation: the system prioritizes the integrity of established rules over rapid intervention, a viable strategy in an environment where existential crises are rare. In practice, this governance style is “muddling through”—a low-cost strategy that only a safe world can afford. Even if the government is inefficient, society runs fine on inertia.

The United States has pushed this style to its extreme. As a “supersized England”, the US also won the geographic jackpot: stable agriculture, secure borders, and vast oceans separating it from old-world strife. Living in this comfort zone for centuries, the United States has developed a highly formalized governance structure where due process is paramount. While this produces a governance culture centered on the rule of law and the protection of established rights and boundaries, it also leads to a more litigious environment where legal deliberation often precedes real-world action. Without the immediate pressure of an existential threat, adherence to procedure becomes the primary mechanism for maintaining social consensus. In such a system, the necessity for broad agreement often leads to what has been termed a vetocracy, where the numerous veto points slow the pace of resource mobilization. This is not a failure of will, but a structural trade-off: a system designed to protect many different interests in a stable environment naturally prioritizes procedural integrity over the speed of action.

In contrast, China faces a highly volatile environment. Amidst floods, droughts, and the social upheaval they triggered, no eloquence can substitute for action; if the state cannot tame the waters or sustain the population, it loses its "Mandate of Heaven." Here, legitimacy is derived not from the process, but from the tangible outcome of survival. Consequently, the delivery of results becomes the primary political ethic. The state cannot be a mere referee; it must be an active builder, judged by its concrete record of development. In this context, order is not a pre-existing condition to be protected, but a physical outcome that must be engineered.

The engineer’s focus on problem-solving thus establishes the physical foundation required for legal certainty to take root, allowing rights to be meaningfully defined and effectively enforced. As we saw with England, stable output anchors prices, lowers enforcement costs, and reduces the transaction costs of social coordination. In essence, China’s Big Push was precisely a massive effort to lower systemic transaction costs that nature had initially set too high. Its success in

agricultural de-risking allowed China to engineer what England had received as a providential gift: the material foundation for legal certainty and self-sustaining growth.

The distinction between preservation and transformation is not a cultural preference but a consequence of the governance costs that the physical environment imposes. In a low-volatility environment, the transaction costs of social coordination are structurally low: disputes are distributional, tax bases are predictable, and public goods demand is modest. In a high-volatility environment, these costs are structurally high: disputes become existential, tax bases are volatile, and demand for disaster prevention, famine relief, and social order maintenance is recurrent and urgent. The governance mode that emerges is one adapted to meeting these costs within the available fiscal capacity.

As engineered stability reduces volatility, however, the governance imperative shifts from transformation toward preservation: defining rights and enforcing contracts. China's reform era reflects precisely this shift. Conversely, as the competitive pressures of a more industrialized world intensify, Western economies face growing demands for rapid infrastructure investment, technological mobilization, and strategic coordination that purely procedural governance struggles to deliver. Both sides are moving: China toward greater legal certainty and market coordination, the West toward greater state capacity for decisive action. The trajectory is convergence. Though shaped by different starting points and historical paths, this convergence is unlikely to produce a single model. What is emerging is a diversity of governance forms within the same feasible set, in which the capacity to engineer material stability and the institutions that sustain legal certainty reinforce each other. The framework predicts this directional shift as a consequence of altered conditions on both sides; the specific political processes, institutional negotiations, and transitional frictions through which convergence unfolds are questions it does not address.

IX. The Soviet Counterpoint: Big Push that Failed to De-risk

The Soviet Union provides a telling contrast; its failure illuminates China's success.

As the first socialist state, the Soviet Union served as the template for China’s system. It demonstrated powerful organizational capacity and launched the original Big Push through central planning. But the result was tragic: until its dissolution, it never solved its agricultural crisis. Both geographic constraints and strategic failures were to blame.

First, although vast, the Soviet Union's core agricultural zones were confined to high latitudes—heat was insufficient, growing seasons were short. Worse, the harsh continental climate meant erratic rainfall, bringing recurrent drought. These constraints kept grain output both low and volatile: in the 1980s, grain output swung violently between 150 million and 210 million tons, with frequent crop failures (FAO, various years; Nove, 1992).

Second, the planned system allowed the Soviet Union to achieve world-renowned feats (satellites in space, waves of tanks), yet agriculture, the survival base, remained fragile. The

Soviet Union launched massive land reclamation campaigns, trying to force a breakthrough in total grain output. But constrained by the harsh climate, these efforts often delivered meager returns despite massive investment. Even more damaging, unlike China's systematic approach, the Soviet Union's agricultural investment was structurally unbalanced:

- Neglect of Infrastructure: extensive use of fertilizer was limited by insufficient irrigation.
- Logistical Failure: inefficient storage and transport systems led to post-harvest losses of up to 20% (FAO, various years; Nove, 1992), tens of millions of tons rotting in fields or in transit.

Unable to provide stable and abundant grain as an anchor for the currency, the Soviet Union remained confined to the limited sphere of the planned system, failing to create the vast, integrated well-functioning market that China eventually engineered. Confined to the administrative coordination that the planned system could sustain, Soviet industry remained inefficient and its goods uncompetitive on the global market.

With chronic food shortages and uncompetitive industrial goods, the Soviet Union, unwilling to bear the political cost of continued austerity, embarked on a path of high-risk external dependence: exporting oil to import grain. Grain imports climbed year by year from the 1970s, reaching 30–40 million tons annually by the 1980s (FAO, various years; Nove, 1992).

But oil prices are highly volatile. This oil-for-grain dependency became the Achilles' heel of Soviet finance. International trade cannot reliably stabilize a nation's food supply. It depends simultaneously on the stability of external purchasing power, the willingness of suppliers to sell, and the competitive pressure of other buyers in the global market. For a nation that has not crossed the material threshold, none is within its control, and any of these links can break under stress. Substituting one source of volatility for another does not solve the binding problem. In 1985–1986, international oil prices collapsed from around $30 to below $10 and hovered at low levels for years (World Bank, various years). Revenue plummeted, but the rigid demand for grain remained. Coupled with the arms race and the war in Afghanistan, the fiscal deficit exploded. This long-term fiscal bleeding fatally weakened the Soviet Union's capacity to sustain its system.

Faced with this fiscal crisis and the chronic failure of collective agriculture, the Soviet Union did, in fact, attempt a gradual agricultural reform structurally analogous to China's Household Responsibility System. In the late 1980s, a series of legislative measures under Gorbachev progressively opened the option of independent family farming for collective farm workers. Very few chose to leave: by the early 1990s, registered private farms accounted for less than 5% of total arable land, with the vast majority of agricultural labor and assets remaining within the collective and state farm system.

This reluctance had deep roots. The Russian village commune (*obshchina*), traceable to the medieval period, had for centuries served as the peasant's primary mechanism of collective risk-pooling: periodically redistributing land strips to equalize exposure, bearing collective liability for taxes, mediating disputes, and organizing communal labor—functions structurally analogous to Chinese kinship networks. After the 1861 emancipation, the tsarist state preserved

this structure, vesting land in the commune rather than in individuals. Stolypin's reforms (1906–1911) attempted to dissolve it, offering peasants consolidated private title and state credit. By the eve of World War I, roughly a quarter of commune households had formally initiated separation, though those who completed full consolidation into independent farms were considerably fewer, with marked regional variation. The pattern of exit was telling: those who left were disproportionately the wealthiest, who could absorb the risk, and the most desperate, who had little left to lose. The middling majority, whose survival most depended on the commune's insurance function, rationally refused (Pallot, 1999). Stalin's forced collectivization destroyed the commune's traditional form, but the *kolkhoz* reproduced its core logic at larger scale: pooled labor, shared risk, and a range of social provisions—housing, clinics, schools—that, though uneven in quality and coverage across periods and ownership types, no individual farm could replicate. When Gorbachev reopened the exit, the calculus was structurally identical to that faced under Stolypin.

The institutional form kept changing; peasant behavior did not. The reason is clear: what persisted was the high volatility of Russian agricultural output. Where independent farming is viable, farmers embrace it—English tenant farmers, American yeoman freeholders, and Chinese peasants after the silent agricultural revolution all demonstrate this. Engineered stability had made independent farming in China a reliable path to betterment; in Soviet Union, no comparable de-risking had occurred, and independence remained a gamble the average household could not afford.

The tragedy of the Soviet Union shows that state capacity to mobilize resources is a necessary condition for Big Push miracles, but not a sufficient one. Success requires that the Big Push solve the binding problem: systemic agricultural de-risking. This demands respect for physical limits and strategic patience. The Soviet failure lay not just in geography, but in strategy. When agricultural breakthroughs stalled, it did not persist in the arduous transformation of its agricultural base. Instead, it prioritized geopolitical ambitions, channeling volatile oil revenues into welfare and an arms race. This unsustainable dependency came at a heavy price: the Soviet Union collapsed, and wealth built over generations was dissipated. Continental exposure made heavy defense spending unavoidable, but the precise trade-off between arms and agriculture was a leadership choice, not a geographic given. As with China, the Soviet planned system did not create the dysfunction of price signals; it inherited an environment in which they had never functioned reliably. But where China's Big Push ultimately resolved the underlying volatility and rendered its own planning system obsolete, the Soviet Union never crossed that threshold.

China's luck lay in its warmer climate, which meant that each unit of agricultural investment yielded far greater returns. But China also showed a strategic persistence that the Soviet Union, facing diminishing returns on its own agricultural investments, ultimately abandoned. Even in the hardest times, China never gave up on long-term investment in water infrastructure and fertilizer production, and built an efficient grain transportation and reserve system, securing its own food supply. Avoiding both external dependency and mortgaging the future is the key to China's resilience.

Not every nation can fully secure its own agricultural base. Those that cannot must rely on international markets or powerful allies. The former offers no guarantee of stability, the latter demands a price in sovereignty. China's achievement was to secure the staple grain supply that anchors its domestic market. It remains reliant on imports for soybeans and other feed crops, a significant but non-existential dependency: if this supply line were severed, diets would narrow but the survival baseline would hold.

Today, Russia, the Soviet Union's principal successor, has largely overcome its agricultural crisis, aided by a warming climate that has extended growing seasons, market reforms, and the adoption of modern agricultural technology. As output has risen and become less prone to crisis-level shortfalls, individual and corporate farming has expanded dramatically.

X. A Complementary Case: The Irish Path

To further refine our understanding of the geographic lottery, we can examine the diverse paths of European industrialization. In the "Goldilocks zone" of southeastern England, the Low Countries, and northern France, favorable endowments converged with stability. Throughout most of the continent, however, these two variables move in lockstep: moving east or south from this core, yields typically decline just as volatility rises. This synchronized shift across Europe's geography obscures the distinct contribution of each factor, making the case of Ireland particularly revealing. The Yangzi Delta possessed endowments comparable to England's yet faced far higher volatility, producing the involution trap at the center of this article's analysis. Ireland presents the mirror decoupling. Its maritime climate delivers volatility as low as southeastern England's, yet its endowments (mediocre grain yields on marginal soils, bog-ridden interiors, and poor internal connectivity) place it well below the northwestern core. Together, these two cases fill the off-diagonal cells of the matrix:

	High Endowments	Low Endowments
Low Volatility	England, Low Countries (stable abundance → rapid takeoff)	Ireland Republic (stable mediocrity → eventual takeoff)
High Volatility	Yangzi Delta China (involution trap)	Sub-Saharan Africa (poverty trap)

The Yangzi Delta has already demonstrated that endowments without stability cannot produce industrialization. Ireland demonstrates the converse: stability without endowments cannot produce rapid industrialization but prevents the economy from falling into a trap, allowing accumulation to continue uninterrupted until favorable external opportunities arrive to accelerate what was already in motion.

Furthermore, Ireland's history provides a direct test of a proposition derived from Coase: under low transaction costs (low volatility), even a severely adverse initial distribution can be corrected through incremental adjustment. Ireland's initial distribution was among the worst in Western Europe: a colonial land settlement that concentrated ownership in a foreign Protestant

ascendancy while legally excluding the Catholic majority from property rights. If any case should have been permanently locked into extractive stagnation by its initial distribution, it was Ireland. That it was not, that it recovered, accumulated, and eventually surpassed its former colonizer in per capita income, is powerful evidence that low volatility, by keeping transaction costs low, permitted the Coasean reallocation that high-volatility environments structurally preclude.

1. The Phenomenon of "Stable Mediocrity"

Historically, Ireland presents a unique path: it was a region that possessed geographic stability, yet struggled with the physical hurdles of transport and grain yield. (by 'stability' we mean low interannual volatility in grain yields under a maritime climate; this does not deny the potato-specific vulnerability that produced the Great Famine.)[24]

While Ireland was often characterized as "poor" relative to the north-west European core, its per capita GDP remained remarkably steady, about one-half to two-thirds of Britain across most of 19th–20th centuries (Bolt and van Zanden, 2020). Within the endowments-volatility framework, this represents "Stable Mediocrity": the low volatility (stability) ensured that Ireland stayed in the economic race over centuries even when faced with external suppression, while its lower output (mediocre grain yields, hostile terrain, and poor internal connectivity) constrained its development, keeping it on a lower plateau compared to its neighbor across the Irish Sea.

2. The North-South Divergence

The internal divergence between the North (Belfast) and the South (Dublin/Cork) during the 19th century illustrates how physical landscapes and institutional niches dictate the material threshold of industrialization, even when regions share the same geographic stability.

Ireland's geographic stability supported the widespread adoption of fixed-rent tenancy, consistent with the pattern observed across stable regions. However, under colonial rule, landlords could for much of the 19th century unilaterally raise rents and evict tenants at will, weakening the developmental function of fixed rent across the island. In the North, this damage was partially offset by colonial political arrangements that afforded Protestant tenants greater security, and by the linen industry, which provided an alternative channel for local capital accumulation. In the South, no such offset existed.

- **The Northern:** Integrated Advantages. Northern grain agriculture was comparable to—and in many districts inferior to—the South's wheat belt, but Belfast possessed a "friendly" physical landscape. Its navigable river systems and solid terrain allowed for the low-cost integration of its hinterland with its ports; this physical openness allowed the

[24] The Great Famine (1845 – 1852) was caused by a specific biological shock—potato late blight—amplified by colonial land policies and extreme dependence on a single crop. These were structural vulnerabilities of poverty and crop composition, not expressions of climate-driven output instability. Blight recurred periodically and recovery required deep adjustment, but that adjustment could proceed cumulatively because Ireland's maritime climate provided a stable baseline on which gains were retained. This distinguishes biological or institutional shocks, however devastating, from the permanent, recurring volatility of the continental East Asian monsoon, which erodes accumulation generation after generation until suppressed by engineering.

North to compensate for its mediocre local yields by accessing external resources. Proximity to British coal and grain markets provided a "virtual" abundance. Crucially, Britain permitted the linen industry to thrive in Ireland as a non-competing sector. In the North, cool, wet conditions and the heavy, often poorly drained drumlin soils made wheat relatively unattractive, while Ulster's climate, water resources and smallholder system supported flax cultivation and the rise of linen. Northern capital found a viable industrial exit, allowing a spontaneous industrial takeoff by 1840.

- **The Southern:** Physical Vulnerability and Institutional Shackle. The South faced a far more challenging physical base. The high-rain environment produced a landscape of lush grass but treacherous mud—fostering a thriving livestock sector while severely limiting grain yields and internal connectivity.
    - **The Cost of Physical Weakness:** The deeper constraint was not middling yields but the small extent of wheat land and, therefore, a limited marketable grain surplus. Southern Ireland's wheat yields were middling by European standards, yet a crop mix dominated by oats and potatoes, combined with bog-ridden, fragmented interiors and high transport costs, curtailed tradable output and the fiscal-military base of the state. This physical reality prevented the consolidation of sufficient state capacity for self-defense, making durable subjugation (often described as 'internal colonization') easier. If the South had possessed a landscape more favorable to grain and transport, it might have lacked its world-class pastoral wealth, but it would have gained the internal cohesion necessary to defend its sovereignty.
    - **The Institutional Shackle:** Once conquered, the South faced a second constraint: Britain's 1699 Woollens Act deliberately constrained Ireland's woollen manufactures by banning direct exports to foreign markets and subjecting shipments to Britain to heavy duties, while leaving exports of raw wool (and yarn) to Britain legal, disproportionately hurting the southern woollen regions. Though formal restrictions eased after 1800, the damage was done: the South had lost the critical window for building industrial capacity, leaving it unable to compete when the market finally opened. The net effect was to limit the highest-value segment of the South's most natural industry and tilt investment toward pastoral provisions (butter, salted beef) rather than deepening a woollen value chain; southern ports increasingly functioned as raw-material conduits, and wealth drained to Britain.

3. Revolution and Redirection in the South

Before the South charted its own course, one transformation had already reshaped the entire island. The transfer of land from the Protestant ascendancy to the Catholic tenantry was accomplished not by revolution but by a half-century of incremental legislation. The 1881 Land Act established fair rent, fixity of tenure, and free sale, extending to the whole island principles that Ulster custom had long recognized informally. The 1903 Wyndham Act financed voluntary buyouts on terms attractive to both sides: landlords received a 12% bonus above market capitalization; tenants repaid through 68.5-year low-interest annuities (Wyndham Land Purchase

Act 1903, 3 Edw. VII, c. 37). By the time of independence in 1922, the bulk of Irish land had changed hands. The new state inherited a society of smallholding owner-occupiers, not dispossessed serfs.

This gradual buyout was feasible only because low volatility underwrote every link in the chain. The government could commit to long-term interest subsidies on a predictable fiscal base. Tenants could service 68-year mortgages because harvests were sufficiently reliable that default remained the exception rather than the norm. Legal contracts held firm because no systemic shock intervened to void them. In high-volatility environments, such gradualism is structurally precluded. Recurrent severe shocks erode the repayment capacity of borrowers, destabilize the fiscal base that funds the subsidy, and periodically reverse the gains of prior reforms, as smallholders who cannot service their debts lose their newly acquired land and revert to tenancy or displacement.

Independence in 1922, then, did not initiate the redistribution of land, which was already largely accomplished, but redirected the flow of surplus. Under colonial rule, the economic surplus generated on Irish-owned land still drained to Britain through trade structures, tax arrangements, and institutional bias. Sovereignty gave Ireland the power to retain and reinvest that surplus domestically. With land already in Irish hands and sovereignty secured, the independent government's first priority was to industrialize what the land already produced reliably. The fiscal base this generated then funded the infrastructure projects that expanded and deepened the domestic market:

- **Industrialize the Surplus:** Ireland did not leap into an economic miracle; it first built the "stomach" to digest its own wealth by industrializing its existing stable production. Agro-processing had deep roots predating independence: dairy cooperatives had spread since the 1890s, and brewing and distilling were already major industries. After independence, these sectors expanded further as retained surplus was reinvested domestically. From the 1920s onward, the state actively expanded this base: meatpacking scaled up, dairy cooperatives consolidated, and world-class brewing and distilling transformed what were once "live exports" into high-value processed goods. Because these industries relied on a steady supply of raw materials, a requirement that Ireland's stability could reliably meet, they provided the foundational industrial coordination and capital accumulation that allowed the nation to build the productive base.
- **Securing the Energy Anchor:** Starting in the 1920s with the Shannon Scheme, a massive hydroelectric project, Ireland began to engineer its own energy sovereignty. This was followed by systematic peat development from the 1930s through the 1960s. The government mechanized peat extraction and commissioned peat-fired power stations, expanding the domestic fuel base. This partly reduced reliance on imported coal and provided a measure of energy security to support industrial growth. Recognizing that the industrial growth of the 1960s and 1970s remained fragile due to a heavy reliance on volatile oil imports, the state launched a decisive wave of energy de-risking in the 1980s. The commissioning of large coal-fired baseload capacity and the arrival of domestic offshore natural gas, later reinforced by a gas interconnector to Britain, helped anchor the power system in more stable, predictable sources. By the 1990s, Ireland had largely

reduced oil exposure in power generation, enhancing the infrastructure foundation required for sustained industrial growth.

- **Integrating the Interior:** The new state inherited a rail network designed to extract resources for Britain: a radial system centered on Dublin and the ports, with weak internal connections. Repurposing this colonial infrastructure took decades. From the 1950s, the focus shifted to road development. Uneconomic rail lines were closed, and road transport, especially motor freight, expanded rapidly. The real transformation came with European funds: from the 1970s, EC regional funds supported highway improvements, and after 1988, EU structural funds enabled large-scale motorway construction. From the 1990s onward, the interior was gradually connected by modern roads, transforming a once-fragmented landscape into an increasingly integrated domestic market.

4. The Modern Leap

Ireland (the Republic of Ireland) did not invent prosperity, it simply removed the physical and institutional dampeners that had long suppressed it. This patient reconstruction of the material base ensured that when the global opportunity, driven by deepening European integration and the surge of footloose high-tech capital, arrived in the 1990s, Ireland's foundation was finally hard enough to support an economic miracle.

Ireland's journey proves that endowments are not destiny, but constraints to be overcome. Unlike nations trapped in highly volatile environments that require a Big Push just to kick-start development, Ireland possessed the stability for persistent accumulation: incremental engineering within a well-functioning market, where each project could build on the last and generate returns over time. By leveraging a stable fiscal base to mobilize domestic and external capital, Ireland transformed itself from a "stable but mediocre" pastoral society into a global industrial frontier. It is the story of a nation that used its "geographic lottery ticket" of stability to rise from subjugation to prosperity.[25]

---

[25] Poland provides an Eastern European mirror of the Irish pattern. Poland's core agricultural regions, situated on the North European Plain, enjoy a climate more continental than Ireland's maritime regime but substantially more stable than the monsoon zones or the Russian interior; geographic stability is moderate. Grain yields were middling by European standards, and the flat, open terrain, while facilitating internal transport, provided no natural barriers against external aggression. This combination of moderate stability and mediocre endowments placed Poland in a situation similar to Ireland's: sufficient stability to sustain institutional arrangements based on fixed claims, but insufficient surplus to build a strong fiscal-military base.

The Polish Sejm was among the most powerful parliaments in early modern Europe. The *liberum veto* granted any single deputy the power to block legislation; the king was elected; the *szlachta* (nobility), comprising roughly 8 to 10 percent of the population, enjoyed a breadth of political participation unmatched elsewhere on the continent. In the early modern period (sixteenth to eighteenth centuries), the *folwark* system was widespread across Poland, with noble-operated demesne farms relying on serf corvée. This is consistent with local output levels still too low to sustain a developed tenancy market: the level of output determines whether tenure is coerced (labor-rent) or market-based. In the nineteenth century, as agricultural modernization raised output levels, the more stable western regions transitioned first to fixed-rent arrangements, while the eastern territories, closer to the volatile continental interior and more sparsely settled, retained labor-rent and manorial structures longer.

Moderate stability made parliamentary governance viable: low volatility minimized the demand for centralized intervention, and decentralized arrangements could persist. But mediocre endowments left the state without the

5. Ireland and India: A Colonial Comparison

A revealing comparative test is Ireland versus India. Both were subjected to prolonged British colonial rule. Both experienced catastrophic famines under that rule: Ireland in the 1840s, India repeatedly through the nineteenth century and into the twentieth. Both gained independence in the mid-twentieth century with deep structural damage: drained surplus, distorted infrastructure, and suppressed domestic industry. Yet their trajectories, both under colonial rule and after independence, diverged fundamentally.

Ireland's per capita GDP, already at roughly half Britain's level at independence, climbed steadily and surpassed Britain's by the early twenty-first century. India's per capita GDP, far lower at independence, improved but remained a fraction of Britain's seven decades later.Divergences of this kind are commonly attributed to differences in governance, social cohesion, or cultural factors. Within the endowments-volatility framework, a simpler and more fundamental explanation is available.

Ireland's low-volatility baseline shaped its trajectory long before independence. Even under colonial extraction, the stability of output ensured that productive capacity, social organization, and human skills were preserved rather than periodically destroyed. By the early twentieth century, land reform had already transferred most holdings to owner-occupiers, and agro-processing cooperatives had taken root, achievements that a high-volatility environment

---

fiscal-military capacity that continental exposure demanded. The liberum veto, an institution perfectly rational for a *szlachta* whose primary concern was protecting landed income from royal appropriation, paralyzed the military mobilization that this exposure required. Poland was partitioned out of existence (1772 – 1795) not because its institutions were economically dysfunctional, but because its surplus base could not sustain the cost of continental defense. Ireland was colonized for structurally identical reasons: insufficient fiscal-military capacity rooted in the same economic mediocrity. But the subsequent trajectories of both demonstrate that colonization and partition do not destroy the foundations of the long-run equilibrium; stability preserves the roots of productive capacity under external domination.

The communist period provides a further test. Poland is virtually unique within the Soviet bloc in that it never completed agricultural collectivization: approximately 75 to 80 percent of arable land remained in private hands throughout the entire period. The government attempted collectivization in the early 1950s but abandoned the effort in the face of sustained peasant resistance. Where independent farming is viable, farmers resist absorption into collective risk-pooling arrangements they do not need. Just as Russian peasants rationally refused to leave the *kolkhoz* under high volatility, Polish peasants equally rationally refused to join one under moderate stability. The stability of agricultural output, sustained by moderate geographic conditions and reinforced by the persistence of private farming, ensured a relatively stable domestic food supply throughout the communist era, sparing Poland the chronic grain crises that plagued the Soviet Union and contributing to an economic performance that ranked in the upper tier of the Eastern bloc.

Post-transition Poland confirms the "stable mediocrity" trajectory. Since 1989, Poland has been the strongest sustained performer among post-communist economies: it was the sole EU member state to avoid recession during the 2008 financial crisis, and its per capita GDP (PPP) has risen to approximately $45,000 by the mid-2020s (World Bank, various years). As with Ireland, the explanation is cumulative: low volatility preserved the foundations of productive capacity, including independent farming, human capital, and social organization, through successive episodes of external domination (partition, occupation). When sovereignty was restored and external constraints lifted, decades of suppressed but never destroyed accumulation could finally express itself.

Together, Poland and Ireland converge from opposite ends of the European continent and under vastly different political histories: stable mediocrity does not produce spontaneous rapid industrialization, but it prevents the economy from falling into a trap. Accumulation is never interrupted, and the foundation for eventual development builds steadily from within.

would have repeatedly undone. After independence, this pattern continued: every increment of investment, in hydroelectric power, in road construction, in education, was retained. Each project built on the last. The gains of the 1930s were still in place in the 1960s; the gains of the 1960s were still in place in the 1990s. No systemic shock intervened to destroy accumulated capital or scatter organized populations. When the external opportunity of European integration and mobile capital arrived in the 1990s, a century of incremental accumulation had prepared a foundation hard enough to support the final breakthrough.

India's high-volatility baseline shaped a very different trajectory, again well before independence. Its agricultural conditions are in many respects more challenging than China's: the South Asian monsoon is more erratic, and intense heat compounds pest pressure. Recurrent shocks periodically destroyed accumulated capital and disrupted social organization, keeping much of the subcontinent locked in a cycle of accumulation and dissipation. Colonial extraction compounded these difficulties but did not create them. After independence, the Green Revolution achieved a genuine but regionally concentrated breakthrough. In Punjab and Haryana, the combination of irrigation, high-yielding varieties, and industrial fertilizer compressed yield volatility enough to support owner-cultivation with heavy fixed investment, long-term bank lending, and fixed cash rent over sharecropping where tenancy existed. Yet across the eastern plains and the rainfed zones of peninsular India, monsoon dependence and high volatility persisted. India's challenge was not that de-risking failed everywhere, but that its success, though sufficient to eliminate famine and achieve aggregate food security, remained too geographically concentrated to release the bulk of the agricultural labor force. The geographic heterogeneity of the subcontinent itself accounts for this concentration: the flat, snow-fed plains of Punjab were physically the easiest to de-risk, while the flood-prone eastern plains and the rainfed peninsular plateau posed far harder physical problems that remained intractable. India pursued the same strategy as China, planned system, heavy-industry priority, and agricultural modernization, but the more severe governance paradox it faced, a thinner surplus base compounded by the absence of a comparable tradition of centralized mobilization, constrained extraction to a lower intensity; it avoided the concentrated catastrophes that China's razor-thin margin produced, only to pay a diffuse and unending price in countless smaller disasters that, generation after generation, prevented the material threshold from being crossed at national scale. The governance paradox persisted at the national scale. The result has been neither collapse nor the broad manufacturing-led transformation that absorbed hundreds of millions into factory employment in China, but a pattern of significant yet uneven progress: output substantially raised, services and urban centers growing rapidly, yet without the structural shift that would draw the bulk of the agricultural labor force into industrial employment.

This comparison eliminates colonial legacy as the decisive variable. Both nations suffered prolonged colonial extraction; what differed was the physical environment in which that extraction occurred and from which recovery had to proceed. Ireland's low volatility was uniform across its small territory, preserving the foundations of recovery even under colonial rule and permitting incremental accumulation everywhere. India's engineered stability was spatially concentrated: its most advanced regions locally overcame volatility, but the national system remained subject to the aggregate instability that made incremental progress insufficient at

continental scale. The divergence is not a judgment on governance quality or cultural capacity. It is a consequence of the physical difficulty of the problem each nation faced.

Policy Implications: A Successful Big Push for the Global South

The perspective of geographic volatility complements rather than replaces the established perspective of geographic endowments. Adverse endowments constrain the speed of development; environmental volatility determines whether development gains can be sustained at all. The stagnation trap of traditional China reveals that high volatility alone can block industrialization even where the level of output is high. In reality, late-developing nations facing systemic underdevelopment typically suffer from both constraints of low output and high volatility simultaneously, making the challenge all the more severe.[26]

Thinking back to the development stories of East and West, we can draw a crucial lesson for the present: underdevelopment is not, at root, a product of cultural traits or bad governance, but a natural result of these severe geographic constraints.

- **West Asia and North Africa:** an inherently water-scarce region. In the modern era of population explosion, this is no longer a problem that can be solved by building a few reservoirs; it is an absolute scarcity.
- **Sub-Saharan Africa:** a region facing multiple geographic curses. Tropical diseases weaken the workforce; rainforest and savanna climates make the threshold for agricultural development exceptionally high; and the lack of navigable rivers makes inland transport prohibitively expensive.

Western economic orthodoxy, accustomed to theorizing from the luxury of a guaranteed harvest, assumes that as long as Western institutions are copied, prosperity will follow. However, this assumption fails to account for the material constraints: in highly volatile environments, institutional blueprints cannot be implemented without first securing a stable physical foundation. Ironically, Western policy practice has long acknowledged what Western economic

---

[26] An intuitive way to grasp the relationship between the two dimensions: the output is the engine; the volatility is the steering. The steering determines whether an economy stays on the road of development; the engine determines how fast it travels. With reliable steering (low volatility), the economy stays on course: a stronger engine (high output) simply means faster progress. This is the difference between England and Ireland. With failed steering (high volatility), the vehicle veers off the road no matter how powerful the engine. This is the involution trap of the Yangtze Delta. When a weak engine coincides with failed steering, the result is the poverty trap: the vehicle is both off the road and unable to move. In practice, engineering that tames volatility and engineering that raises the output are often inseparable facets of the same Big Push. A reservoir both prevents floods and irrigates new cropland; a road both opens markets and enables emergency relief; a power grid both drives production and powers drainage pumps in a flood. But de-risking is the binding constraint: without it, no amount of additional capacity can deliver self-sustaining growth. A weak engine alone does not prevent transition: population tends to equilibrate with the available food supply, so even modest output, if stable, can sustain a well-functioning market, as Ireland demonstrates. Yet stability is a platform, not a launch; the transition to industrialization requires rising output on top of stability, releasing labor and capital from agriculture. England's agricultural revolution took off as output increased through improved rotations and land consolidation; China's transition began in the early 1980s when per capita grain output, still a modest 320 kg, was simultaneously stabilizing and rising, surging from about 305 million to over 400 million tons by 1984 as fertilizer investments matured and household incentives took effect (National Bureau of Statistics of China, various years). The Soviet comparison sharpens the point. Regardless of the precise level of per capita output, Soviet grain production remained chronically volatile, swinging between roughly 150 million and 210 million tons across the 1980s, with substantial post-harvest losses further eroding the usable supply. It was this instability that blocked the transition to a well-functioning market.

theory denies. American policymakers designing Cold War assistance, from the Marshall Plan in Europe to PL-480 'Food for Peace' programs spanning Asia, the Middle East, and Latin America, understood intuitively that without stable food supply there is no social order, without social order there is no free market or rule of law, and without these there is no bulwark against revolution. They intervened at the beginning of the causal chain, and stability, order, functioning markets, and legal certainty followed.

This article makes explicit what practitioners have long known in practice. Among economists, Jeffrey Sachs has come closest to translating this practitioner knowledge into academic prescription. As he has pointed out, for nations trapped in poverty, standard market prescriptions simply do not work; they need a Big Push—massive coordinated investment in infrastructure and agricultural inputs. This diagnosis has struggled to gain traction against alternative approaches that focus on marginal interventions within the existing environment rather than transforming the environment itself. Microfinance seeks to unlock individual entrepreneurship through small loans; randomized controlled trials test the impact of targeted provisions such as bed nets or fertilizer subsidies; property titling aims to convert informal assets into collateral. Each addresses a real constraint at the individual level. But from a systemic perspective, they share a common blind spot: none of them alters the systemic volatility that periodically destroys the gains they enable. In an environment where a single drought can wipe out a season's earnings, a small loan cannot build a durable enterprise; a bag of fertilizer cannot compensate for the absence of irrigation; and a property title cannot hold its value when output is chronically volatile. They may bring temporary relief, but in a systemically volatile environment, such gains are written in sand, erased by the next shock before they can accumulate. These are not failed ideas but incomplete ones: they operate within the constraints that only systemic engineering can remove.

Even when the need for a Big Push is recognized, the Chinese and Soviet experiences carry a sobering lesson: a Big Push that extracts surplus from a fragile agricultural base operates on a razor-thin margin of error. It is an inherently arduous and dangerous path. China narrowly succeeded; the Soviet Union did not. Moreover, most nations of the Global South could not replicate it even if they wished to. They face an even more severe version of the governance paradox: the very conditions that demand a Big Push also undermine the state capacity to initiate and sustain one. Where China could mobilize a continent-spanning bureaucracy forged over millennia of coping with volatility, and possessed, however precariously, an agricultural surplus large enough to sustain a self-reliant Big Push, many of these nations lack not only the administrative reach, fiscal base, or social cohesion to undertake extraction at such scale, but also the agricultural surplus from which to extract.

In principle, a self-reliant Big Push is not the only path to industrialization: high-volatility regions can also be pulled into industrialization by merging into a low-volatility core's system, as the East Asian economies merged into America's. This is itself a form of Big Push, but one whose costs are borne by the external core rather than extracted from the nation's own fragile agricultural base. It requires a scale small enough for the guarantor to absorb, and a geopolitical willingness to bear the cost—conditions unavailable to most nations of the Global South today.

Fortunately, today's toolkit is fundamentally different. The geopolitical rivalries that once motivated great-power sponsorship of small allies have largely receded, but in their place, international cooperation and technological capacity have both expanded dramatically. Agricultural technology has advanced dramatically: drought-resistant crop varieties, precision irrigation, solar-powered pumping, and satellite-based weather forecasting can deliver de-risking at far lower cost and greater speed than the brute-force engineering of the mid-twentieth century. The international environment has also improved: global agricultural research networks, multilateral development finance, and the diffusion of Chinese manufacturing capacity have made critical inputs more accessible than ever. The challenge for the Global South is not to repeat China's arduous path, but to find a safer one, leveraging modern technology and international partnership to cross the material threshold. For nations caught in the governance paradox, such external partnership is not merely helpful but indispensable.

Infrastructure and agricultural investment in the Global South, including but not limited to China's Belt and Road Initiative, resonates with the logic of Sachs's 'clinical economics.' By investing in transport and energy infrastructure, and by transferring agricultural technology such as hybrid rice and irrigation techniques, such partnerships are helping build the material foundation that must precede a well-functioning market. Though this may appear to have little to do with institutions, it is precisely what transforms the baseline institution, and that is the institutional prescription that truly works. The engagement of industrialized partners with the Global South through trade and infrastructure investment is consistent with the sequence that this article has identified: build the foundation first, and prosperity follows.

Yet this is only the very beginning. A successful Big Push for the Global South will require going beyond current efforts to address the deeper constraints of food and water security, with far more coordinated planning and far greater resource commitment than what has been mobilized so far. Such commitment is justified on both moral and practical grounds: as more nations overcome geographic constraints and build well-functioning markets, they join as new participants who expand the global market and generate gains for all.

Conclusion: The Renewal of Civilization

The story of modernization is not the triumph of one culture over another, but a testament to humanity's capacity to master its own material environment. It proves that poverty and stagnation are not destinies dictated by the physical landscape. Even in the most volatile environments, the material threshold of industrialization can be crossed through a Big Push that overcomes the binding constraint: systemic agricultural de-risking. By constructing a stable material foundation, human society can finally break the shackles of geographic constraints.

For millennia, civilizations were trapped in scarcity and unpredictability. Because the material world was volatile, surpluses could not be sustained, and competition for limited resources often turned violent, leading to a global order anchored in plunder and war. Systemic engineering is breaking this cycle. When humanity uses systemic engineering to stabilize the supply of food and energy, human cooperation shifts from a struggle for survival to a positive-sum game of continuous accumulation. This creates the first real opportunity to move beyond the defensive logic of the old era. The focus of governance shifts from prescribing institutional forms to constructing material foundations, and the world order is moving toward a new era where building replaces snatching.

This journey echoes the ancient narrative of Jacob wrestling with the divine through the long night until the break of dawn. In the pre-modern era, humanity was "Jacob", a wanderer at the mercy of an unpredictable Providence, struggling against the chaos of a volatile nature. By achieving material predictability through engineering, we are realizing the deeper meaning of the name "Israel": those who wrestle with fate and prevail.

China's experience is but one manifestation of this universal trajectory. Humanity is moving toward a shared future where engineered stability replaces geographic chance as the bedrock of order. We are building a world where human dignity is no longer a providential luxury, but an engineered baseline secured for all.

Appendices

Appendix A: Why Geographic Volatility Is the Binding Constraint

Geographic volatility is critically binding because it shapes the reliability of the price signals and contract enforcement on which the market mechanism depends. It recurs with every annual harvest, affects every person who must eat, and threatens the survival baseline on which all economic activity depends. No other source of volatility shares this universality: geographic volatility has operated continuously across every agricultural civilization in human history, from the earliest settled communities to the eve of modern engineering. War, epidemic, and commercial shocks, however devastating in particular times and places, are episodic or localized; geographic volatility alone is permanently present wherever humans depend on the harvest.

A brief enumeration clarifies the point. War can be existentially grave, but it is episodic: even the most war-torn regions experience intervals of peace during which trade and exchange recover. Epidemic disease is similarly episodic: the Black Death devastated Europe for several years and then receded. Endemic tropical disease (e.g., malaria) is persistent within affected zones, but it is geographically localized rather than a universal constraint on all agricultural civilizations, and its economic effect operates through the mean level of output, reduced labor productivity and shortened life expectancy, rather than through interannual volatility; its impact on the market mechanism is not significant. Commercial volatility and financial volatility in pre-modern societies were largely endogenous to food-supply shocks: grain-price spikes triggered hoarding, credit contraction, and the collapse of purchasing power for non-food goods. They transmitted and amplified agricultural volatility rather than constituting an independent source of volatility in their own right.

Appendix B: North's Beliefs, Path Dependence, and Material Conditions

What Douglass North called 'beliefs' or 'mental models'—the culturally inherited priors that he treated as independent drivers of institutional choice — are better understood, within this framework, as primarily shaped by the physical environment rather than as independent priors that autonomously drive institutional choice. The Dujiangyan case is dispositive. Where ancient irrigation engineering virtually eliminated output volatility, agents inside the stability zone, within the same culture, legal tradition, and political system, adopted rigid fixed-rent contracts and responded as residual claimants with incentives to invest and accumulate, while those outside it remained locked in risk-sharing arrangements characteristic of high-volatility environments. What changed here was not belief but volatility.

This reframing also resolves two further Northian concepts. His "credible commitment" (North and Weingast, 1989), the willingness to honor long-term obligations, presupposes the stable surplus that low volatility provides; it is the economic expression of what Mencius called "a

steady mind born of a secure livelihood." His 'path dependence,' within this framework, is primarily driven by the persistence of the material conditions of surplus: institutions endure not because they are locked in, but because the conditions that selected them endure. Once those conditions are altered, as China's agricultural engineering demonstrated, institutions shift accordingly. The transitional inertia North identified (learning effects, coordination costs, adaptive expectations) may slow adjustment, but the direction of change confirms the primacy of material conditions.

Appendix C: Reinterpreting the Stiglitz Sharecropping Model

The Stiglitz (1974) model assumes that effort is unobservable by the landlord and, within a principal-agent framework, establishes that sharecropping is a second-best optimal contract under this information constraint. The model then derives comparative statics showing how the optimal share ratio depends on three core parameters: the variance of output, the degree of tenant risk aversion, and the marginal contribution of effort to output. This formalization was historically important. Marshall had argued that sharecropping is necessarily inefficient because the tenant receives only a fraction of the marginal product, leading to suboptimal effort. Cheung (1969) challenged this verdict from the perspective of transaction costs. Stiglitz's distinct contribution was to accept that the incentive distortion is real, but to demonstrate, within the language of information economics, that sharecropping is nevertheless optimal when volatility is sufficiently high relative to the incentive cost: the gain from risk-sharing outweighs the loss from weakened incentives. As a general principal-agent model, the separation of these parameters is logically sound. However, when the model is brought back to the empirical conditions of its original motivating application, pre-modern agriculture, the apparent independence of these parameters proves to be an artifact of the formalization, not a feature of reality. Under the empirical conditions of pre-modern agriculture, the foundational assumption and all three parameters collapse into downstream consequences of a single variable: the magnitude of output volatility imposed by the physical environment. (In pre-modern agriculture, high variance and severe downside asymmetry are empirically bundled; "volatility" here carries this compound meaning.)

Risk aversion is a universal human psychology, not a parameter that varies across regions. Cross-cultural experiments measuring risk attitudes (Falk et al., 2018) do find systematic differences across populations, but these differences are broadly consistent with the agricultural volatility that the physical environments of their regions of origin impose. This volatility is not a historical relic: the climatic regimes that generated it persist today, and the engineering that suppresses it, principally the Green Revolution package of irrigation, fertilizer, and improved seeds, has been in place for barely two generations and remains incomplete in many regions. Populations shaped by generations of high-volatility agriculture, a condition that in most cases has only recently begun to be mitigated, exhibit higher measured risk aversion. Cultural attitudes toward risk, what North called "beliefs" or "mental models," are themselves downstream products of the physical environment rather than independent priors. Landlords and tenants in England do not differ in underlying psychological aversion to risk from landlords and tenants in the Yangtze Delta; what differs is the magnitude of the risk their environment imposes. Since risk

aversion is effectively constant across populations, the operationally varying factor is volatility alone. High output volatility directly makes fixed rent unacceptable to tenants regardless of their wealth. Disaster years recur with sufficient frequency that a rigid fixed payment entails a high probability of severe economic loss or outright insolvency. Even a tenant with substantial reserves would rationally refuse this contract, because the expected losses in bad years are large enough to outweigh the incentive advantages of residual claiming in good years. When disaster years are frequent and severe enough, the highest fixed rent the tenant can reliably service falls below the landlord's expected revenue under sharecropping; no mutually acceptable fixed rent exists. In a low-volatility environment, the risk is small enough that it can be compensated by a predictable reward: the tenant gains the full residual claim on all surplus above the fixed rent, and can expect to collect it reliably year after year. Tenants are no less averse to risk, but the risk is small and the return is dependable, making fixed rent a rational choice. The same logic operates on the landlord's side. In a low-volatility environment, tenant default is rare, so the landlord receives a near-certain income stream. The upside forgone by not sharing in bumper harvests is modest, because the gap between good years and normal years is narrow. Meanwhile, the tenant's complete incentive to maximize output raises total production above the level attainable under sharecropping, where Marshallian incentive distortion suppresses effort; in a competitive tenant market, the landlord captures part of this efficiency gain through a higher fixed rent, while saving on monitoring costs (Cheung, 1969). In a high-volatility environment, the landlord's calculus reverses. This point extends beyond Stiglitz's formal model into the empirical reality of contract enforcement: attempting to enforce a fixed payment during a major harvest failure produces not wealth but social friction, disputes, tenant flight, and the cost of finding replacements. The enforcement cost of fixed rent becomes prohibitive, and the landlord is compelled to accept sharecropping. Thus for both parties, it is the magnitude of volatility that determines which contract is rational.

The marginal contribution of effort to output, which the model treats as an independent production parameter, is in practice governed by volatility through the downside asymmetry that is empirically bundled with high variance in pre-modern agriculture. In catastrophic states such as basin-wide floods or severe droughts, effort yields zero return regardless of its intensity. But the effect is not limited to such extremes: even in non-catastrophic bad years, adverse conditions such as waterlogging, drought stress, or pest outbreaks reduce the physical effectiveness of labor inputs, so that the same effort produces less additional output than it would under normal conditions. The expected marginal product of effort therefore declines as the frequency and severity of adverse states increase (Rosenzweig and Binswanger, 1993; Morduch, 1995). What varies across environments is not the intrinsic diligence of farmers but the expected payoff of their effort, a payoff that the compound risk profile of volatility determines.

The model's foundational assumption itself, that effort is unobservable, is likewise a consequence of volatility rather than an independent institutional feature. As volatility rises, output becomes an increasingly unreliable signal of effort: the noise of nature progressively drowns the signal of diligence, and the precision with which the landlord can infer effort from observed output declines continuously. In a low-volatility environment, this signal is sharp enough that high output almost certainly reflects diligence and low output almost certainly reflects shirking,

making output a reliable proxy for direct observation. As volatility increases, this inference degrades until, at high levels, output ceases to be informative about effort altogether. If volatility were zero, effort would be perfectly revealed by output alone, and the entire model would collapse, because there would be no information problem to solve. The assumption of unobservability is therefore not a separate premise but a description of where a given environment falls on the continuous spectrum of signal precision, a spectrum that volatility itself determines.

The effective degrees of freedom explaining contract choice in pre-modern agriculture therefore reduce to one: the magnitude of output volatility. This does not invalidate Stiglitz's formal results; within the model's own terms, the derivations are correct. But when applied to the empirical domain that motivated it, the explanatory power derives entirely from output volatility, not from the interplay of independent forces. The formalization, while important for settling the Marshallian debate within the discipline, inadvertently obscured the simplicity of the underlying insight: where output is volatile, sharecropping emerges; where output is stable, fixed rent prevails.

The foregoing analysis addresses the choice between fixed rent and sharecropping, both of which presuppose output sufficient to sustain market-based tenancy. Below this threshold, where local output is too low to support the operation of a tenancy market, the landlord cannot reliably extract rent through market mechanisms and must resort to direct coercion: labor-rent, in which the tenant renders labor on the lord's demesne in exchange for access to subsistence land. Historically, labor-rent prevailed universally in regions where local markets were undeveloped—early medieval Western Europe being the general case. The Medieval Warm Period raised output levels, local markets gradually developed, and the dissolution of labor-rent accompanied this process. Labor-rent is thus governed by the endowments dimension of the framework rather than the volatility dimension. The three canonical tenure forms map onto the two-dimensional structure: the level of output determines whether tenancy is coerced (labor-rent) or market-based; conditional on market-based tenancy, the volatility of output determines whether the contract is rigid (fixed rent) or flexible (sharecropping).

Appendix D: State Formation, the Leviathan Typology, and Geographic Stability

The influential literature on European state formation in historical sociology and institutional economics (Tilly, 1990; North and Weingast, 1989; Acemoglu and Robinson, 2019) argues that interstate military competition drove the emergence of parliamentary constraints and property rights protection in Europe. Yet the core cases in which this process produced durable parliamentary constraints and sustained economic development are concentrated in the same low-volatility zone this article identifies: England, the Low Countries, and to a lesser extent northern France. Where geographic stability kept the stakes of conflict limited and the fiscal base predictable, military competition could be resolved through negotiated compromise rather than absolute coercion. Traditional China provides the clearest countertest: military competition was

no less intense, and output no less abundant, yet the outcome was centralized authority rather than parliamentary constraint, because high volatility frequently escalated fiscal and military crises to existential severity, structurally favoring centralized mobilization over negotiated compromise. This suggests that what the literature attributes to war is better understood as a joint product of war and geographic stability.

The same material logic determines the type of Leviathan a society can sustain. Acemoglu and Robinson's The Narrow Corridor classifies Leviathans as Shackled, Despotic, or Absent, defined by the balance of power between state and society. This typology accurately describes the phenotype of governance, but leaves the genotype unidentified: it does not explain why some societies achieve balance while others do not. Within the endowments-volatility framework, the answer is straightforward. The Shackled Leviathan is the institutional expression of a low-volatility environment: stable output minimizes the demand for state intervention, so the state can govern without expanding, and society can restrain it without confrontation. As volatility rises and the output remains sufficient to sustain a state, governance is progressively forced toward coercive extraction: volatility simultaneously inflates the demand for state action and erodes the fiscal base, producing the Despotic Leviathan. At the extreme, where high volatility coincides with output too low to sustain even minimal state capacity, the governance paradox becomes unresolvable: the state cannot provide public goods, producing the Absent or Paper Leviathan. The 'narrow corridor' may be a political achievement; but low volatility is the material condition under which the corridor exists at all.

England did not arrive at constrained governance by design. In a low-volatility environment where existential crises were rare, concentrated royal authority cost more than it was worth, and Parliament progressively curtailed traditional royal prerogatives. When the Stuarts resisted, the nation paid the price of civil war, confirming that the environment had already tilted the balance. Through costly political experience, actors converged on limited government as the lower-cost equilibrium.

It should be noted that the criteria Acemoglu and Robinson employ to identify the "Shackled Leviathan" are not robust. They tend to treat observed power-sharing structures, such as Botswana's kgotla (tribal assembly), as direct evidence of institutional constraints on state power. Yet power dispersion can arise from at least two fundamentally different causes: one in which output is sufficient to sustain centralization but is institutionally prevented, and another in which output is simply too low for anyone to concentrate power in the first place. The latter is formally indistinguishable from the former, yet it reflects the absence of state capacity rather than the presence of institutional constraints on state power.

Numerous historical cases demonstrate that such "pseudo-shackling" collapses as soon as output conditions change. The Mongol kurultai, originally a substantive check among aristocratic factions, began losing its constraining function as Chinggis Khan consolidated power during the unification of the steppe, a process further entrenched by the wealth flowing from the subsequent conquest of settled agricultural territories. In precolonial West Africa, smaller-scale kingdoms and chiefdom confederations were transformed into large centralized empires once trans-Saharan trade

provided the fiscal resources to sustain standing armies and administrative hierarchies, as in the emergence of Mali and Songhai.

The symmetric process is equally instructive. Pre-Norman England had already developed considerable state capacity through a gradual process of political consolidation that did not amount to centralization. The Norman Conquest of 1066 imposed a qualitatively different regime: a conflict-driven concentration of power that replaced the native elite wholesale and placed all land under direct royal authority. Yet in a low-volatility environment, this exogenous centralization was progressively moderated. The common law tradition itself, customary, uncodified, and slow-evolving, could only have taken root and persisted within the feasible set permitted by low-volatility conditions; in a high-volatility environment, such a loose juridical order would have been swept aside by the administrative demands of recurrent crisis. The long trajectory from Magna Carta to parliamentary governance was not an accidental institutional invention but a sustained reversion toward the equilibrium that underlying conditions could support.

China's trajectory confirms the same logic from the high-volatility side. As a high-output, high-volatility environment, China periodically experienced phases of relative political relaxation or regional autonomy, yet recurrent crises—floods, droughts, famines, and the social upheaval they triggered—repeatedly regenerated the demand for centralized mobilization, pulling governance back toward concentration. Where England's low volatility progressively dissolved imposed centralization, China's high volatility progressively dissolved any nascent constraint on state power. The reform era provides the decisive contemporary confirmation: as engineered stability has progressively reduced volatility, governance has shifted measurably toward market coordination and legal certainty, with the balance between transformation and preservation progressively adjusting as material conditions stabilize. This is precisely the direction that altered material conditions predict. To classify China as remaining outside the "corridor" on the basis of its government form is to commit the same identification error in reverse: mistaking phenotype for genotype, while the underlying conditions have already shifted.

Taken together, these cases suggest that what determines the long-run trajectory of a power structure is not its observable institutional form at any given moment, but the underlying conditions of output and volatility. England's post-Conquest reversion and China's post-engineering transition are mirror images of the same process: conditions set the equilibrium, and governance converges upon it regardless of the starting point. A polity may exhibit power-sharing arrangements for entirely different reasons: its institutions may genuinely constrain the state, or material scarcity may simply leave no alternative. In low-output, high-volatility environments, decentralized structures reflect weak state capacity rather than institutional constraints on strong state capacity. This is precisely what makes the identification problem acute: the kgotla looks the same whether it restrains a powerful state or simply fills the vacuum left by the absence of one. To classify both cases as falling within the "corridor" is to conflate fundamentally different causal mechanisms. The condition fundamentally incompatible with limited government is not low output per se, but high volatility layered on top of high output: abundant surplus makes centralization materially feasible while recurring crises

continuously generate demand for expanding state authority into ever wider domains. A high-output, low-volatility environment produces the opposite outcome: sufficient output provides the material basis for effective state capacity, while the prolonged absence of crises removes the justification for expanding that capacity without bound, and the resulting equilibrium is a government that is capable yet limited—whether that limitation takes the form of English parliamentary constraint or some other constitutional settlement.

A crucial distinction follows for nations that have engineered their stability rather than inherited it. In naturally low-volatility environments, the maintenance costs of market coordination and legal certainty are structurally low; limited government suffices because nature itself underwrites the foundation. In environments where stability is an engineering achievement, that foundation must be continuously maintained: reservoirs, irrigation networks, power grids, grain reserve systems, and the industrial base that supplies them all require sustained, large-scale, coordinated state action. Centralized governance in this context is not expansive intervention in the economy but the maintenance of the engineered infrastructure on which market coordination and legal certainty depend—a task that is structurally light where nature provides stability and structurally heavy where engineering must substitute for nature. The two paths converge not on the same form of government but on the same order: market coordination and legal certainty. What differs is the scale of state capacity required to sustain it.

Appendix E: The Dialectic of the Planned System

The transition from the planned system to the market system was not an external correction imposed upon a failed experiment; it was the internal dialectic of the planned system fulfilling, and thereby negating, its own reason for being. The logic unfolds in three steps. First, high geographic volatility rendered grain supply chronically insufficient and unstable, paralyzing market coordination. The planned system arose as a rational adaptation to this reality: the state rationed scarce grain to anchor the currency and carved out an insulated industrial sphere shielded from external turbulence. Second, within this sphere, the planned system mobilized organizational capacity and resources to engineer the very stability that nature had denied—reservoirs, fertilizer plants, improved seeds, and rural electrification systematically suppressed the output volatility that had plagued Chinese agriculture for millennia. Third, once agricultural output crossed the material threshold, becoming both abundant and stable, the conditions that had made the plan necessary ceased to exist. A well-functioning market could now take root on engineered stability. The planned system negated itself, not through failure, but through success.

This dialectic reveals a deeper truth about the ontological status of institutions. Institutions are neither passive reflections of material conditions nor autonomous forces that independently determine outcomes; they are generative intermediaries with real causal efficacy. To say they are not passive reflections is to recognize that the planned system actively mobilized resources to overcome constraints that the same geography, absent this institutional form, had failed to break for millennia, the stagnation of the late Qing and Republican eras is proof enough. To say they are not autonomous forces is to recognize that an institution's validity is bound to the specific material conditions it addresses; once it has altered those conditions, persisting unchanged

converts rational adaptation into institutional inertia.

This also offers an alternative reading of Douglass North's path dependence. Institutions persist not because they are locked in by increasing returns and switching costs, but because the material conditions that selected them persist. When those conditions are fundamentally altered, as they were by the silent agricultural revolution, institutional transformation becomes not merely possible but irresistible. The most effective reforms of the 1980s were not designed from above; they grew from below once the material foundation was in place. When agricultural output became reliably abundant, household farming, market exchange, and industrial entrepreneurship emerged as spontaneous responses to a new reality.

Appendix F: Alternative Paths Assessed

Japan, though geographically East Asian, is climatically closer to the Western pattern. As a maritime monsoon country, its oceanic surroundings moderate the rainfall variability and temperature extremes that define continental China's climate. At the domain level, the Tokugawa *nengu* (annual tribute) was assessed on a nominally fixed *kokudaka* (rice-yield assessment in *koku*), often unadjusted for decades. In practice, however, the rigidity of assessment diverged systematically between west and east: western domains, where output was stable and agricultural development mature, maintained genuinely fixed assessments and predictable collection; eastern and northern domains, where extensive land reclamation and higher climatic volatility produced large gaps between nominal and actual output, frequently reassessed yields internally and adjusted collection according to harvest conditions. And at the tenant level, the same pattern held: fixed-rent contracts predominated in western Japan, while share tenancy was more common in eastern Japan, including the productive Kanto plain, and in the colder northeast. The eastern pattern is particularly telling: the Kanto plain's yields were comparable to the west's, but its greater climatic variability favored risk-sharing arrangements, paralleling the Yangtze Delta case where high output coexisted with sharecropping. The system as a whole incorporated more flexibility than England's, as harvest-failure abatements and collective negotiation were not uncommon, but remained substantially more rigid than the near-routine rent reductions characteristic of continental China. Its low-volatility western regions, centered on the Osaka-Kobe commercial belt, provided the stable agricultural surplus and accumulated merchant capital on which Meiji industrialization could draw, and industrialized earliest. The east, despite comparable yields, lagged behind under higher climatic variability, while the northeast, with shorter growing seasons, lower yields, and higher interannual volatility driven by cold-ocean wind patterns, remained agricultural and impoverished well into the twentieth century.

South Korea and Taiwan, also frequently cited as counterexamples, do not constitute relevant parallels. Both received significant agricultural investments during the Japanese colonial period, though the depth differed: in Taiwan, large-scale irrigation, improved rice varieties, and modern extension services had meaningfully reduced agricultural volatility before the war; in Korea, colonial-era improvements laid a foundation but nationwide agricultural stabilization came mainly in the 1960s and 1970s through sustained investment in irrigation, fertilizer, and improved varieties, proceeding in parallel with export-oriented industrialization. In both cases, postwar

American aid and land reform reinforced the base. Their compact size allowed external intervention to achieve rapid, comprehensive coverage, a path that becomes prohibitively costly and slow at continental scale.

More broadly, no middle path between self-reliant heavy industry and full trade dependence has succeeded at continental scale; none achieved durable and comprehensive self-sustaining growth. Latin American import-substitution industrialization (Brazil, Mexico, Argentina, 1950s to 1970s) financed tariff-protected domestic manufacturing through agricultural and mineral exports; all remained vulnerable to terms-of-trade shocks and collapsed into the debt crises of the 1980s. India's mixed strategy, heavy-industry priority at lower extraction intensity, with a larger private sector and partial trade openness, failed to cross the material threshold at national scale. At smaller scale, Thailand and Malaysia achieved partial success through a lighter version of the same pattern that sustained South Korea and Taiwan: Cold War alliance security, sustained foreign aid, and Japanese FDI, combined with natural or engineered agricultural advantages in their core regions, while their un-de-risked peripheries remained in poverty. Nor does light-industry export offer a viable alternative at scale: garment and similar industries operate on margins too thin (typically 3 to 5 percent) to fund continental-scale agricultural infrastructure, and their revenues depend on external consumer demand while the agricultural de-risking they are meant to fund depends on imported inputs, exposing the country to disruption on both the selling and the buying side. For a continental-scale economy under Cold War isolation, no external stabilizer of sufficient capacity existed, and every trade-based path substituted one source of volatility for another without resolving the binding constraint. The absence of a successful middle path at this scale is not a logical impossibility but an empirical regularity that the framework accounts for: the very conditions that make a middle path attractive, insufficient domestic capacity and a volatile fiscal base, are the conditions that make it unviable.

References

Acemoglu, D. and Robinson, J.A. (2019). The Narrow Corridor: States, Societies, and the Fate of Liberty. New York: Penguin Press.

Allen, R.C. (2009). The British Industrial Revolution in Global Perspective. Cambridge: Cambridge University Press.

Cheung, S.N.S. (1969). The Theory of Share Tenancy. Chicago: University of Chicago Press.

de Vries, J. (2008). The Industrious Revolution: Consumer Behavior and the Household Economy, 1650 to the Present. Cambridge: Cambridge University Press.

Diamond, J. (1997). Guns, Germs, and Steel: The Fates of Human Societies. New York: W.W. Norton.

Falk, A., Becker, A., Dohmen, T., Enke, B., Huffman, D. and Sunde, U. (2018). Global Evidence on Economic Preferences. Quarterly Journal of Economics, 133(4), 1645–1692.

Hayek, F.A. (1945). The Use of Knowledge in Society. American Economic Review, 35(4), 519–530.

Huang, P.C.C. (1985). The Peasant Economy and Social Change in North China. Stanford: Stanford University Press.

Huang, P.C.C. (1990). The Peasant Family and Rural Development in the Yangzi Delta, 1350–1988. Stanford: Stanford University Press.

Li, B. (2000).Agricultural Development in the Yangzi Delta, 1620-1850. Houndmills, England: The Macmillan Press Ltd.; New York, USA: St. Martin's Press, Inc.

Lin, J.Y., Cai, F., and Li, Z. (1996). The China Miracle: Development Strategy and Economic Reform. Hong Kong: The Chinese University Press.

Marshall, A. (1890). Principles of Economics. London: Macmillan.

Marx, K. (1867). Das Kapital, Vol. I.

Mendels, F.F. (1972). Proto-Industrialization: The First Phase of the Industrialization Process. Journal of Economic History, 32(1), 241–261.

Morduch, J. (1995). Income Smoothing and Consumption Smoothing. Journal of Economic Perspectives, 9(3), 103–114.

North, D.C. (1990). Institutions, Institutional Change and Economic Performance. Cambridge: Cambridge University Press.

North, D.C. (2005). Understanding the Process of Economic Change. Princeton: Princeton University Press.

North, D.C. and Weingast, B.R. (1989). Constitutions and Commitment: The Evolution of Institutions Governing Public Choice in Seventeenth-Century England. Journal of Economic History, 49(4), 803–832.

Overton, M. (1996). Agricultural Revolution in England: The Transformation of the Agrarian Economy, 1500–1850. Cambridge: Cambridge University Press.

Pomeranz, K. (2000). The Great Divergence: China, Europe, and the Making of the Modern World Economy. Princeton: Princeton University Press.

Rosenzweig, M.R. and Binswanger, H.P. (1993). Wealth, Weather Risk and the Composition and Profitability of Agricultural Investments. Economic Journal, 103(416), 56–78.

Skinner, G.W. (1964–65). Marketing and Social Structure in Rural China. Journal of Asian Studies,

24(1–3).
Slicher van Bath, B.H. (1963). The Agrarian History of Western Europe, A.D. 500–1850. London: Edward Arnold.
Smith, A. (1776). An Inquiry into the Nature and Causes of the Wealth of Nations.
Stiglitz, J.E. (1974). Incentives and Risk Sharing in Sharecropping. Review of Economic Studies, 41(2), 219–255.
Tawney, R.H. (1932). Land and Labour in China. London: George Allen & Unwin.
Tilly, C. (1990). Coercion, Capital, and European States, AD 990–1990. Cambridge, MA: Blackwell.
Townsend, R.M. (1994). Risk and Insurance in Village India. Econometrica, 62(3), 539–591.
Wong, R.B. (1997). China Transformed: Historical Change and the Limits of European Experience. Ithaca: Cornell University Press.
Yang, X. (2001). Economics: New Classical versus Neoclassical Frameworks. Malden, MA: Blackwell.
Yang, X. and Ng, Y.-K. (1993). Specialization and Economic Organization: A New Classical Microeconomic Framework. Amsterdam: North-Holland.